

UNIT-II

ORTHOGRAPHIC PROJECTIONS

Projection theory

In engineering, 3-dimensional objects and structures are represented graphically on a 2-dimensional media. The act of obtaining the image of an object is termed “projection”. The image obtained by projection is known as a “view”. A simple projection system is shown in figure 1.

All projection theory are based on two variables:

- Line of sight
- Plane of projection.

Plane of Projection

A plane of projection (i.e, an image or picture plane) is an imaginary flat plane upon which the image created by the line of sight is projected. The image is produced by connecting the points where the lines of sight pierce the projection plane. In effect, 3-D object is transformed into a 2-D representation, also called projections. The paper or computer screen on which a drawing is created is a plane of projection.

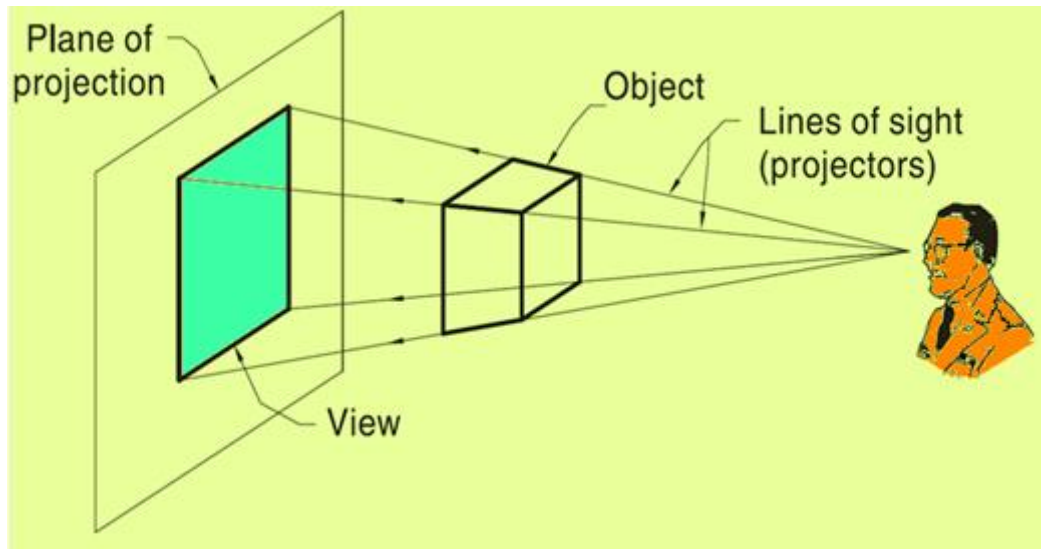


Figure 1: A simple Projection system

Projection Methods

- Projection methods are very important techniques in engineering drawing.

Two projection methods used are:

- Perspective and
- Parallel

Figure 2 shows a photograph of a series of building and this view represents a perspective projection on to the camera. The observer is assumed to be stationed at finite distance from the object. The height of the buildings appears to be reducing as we move away from the observer. In perspective projection, all lines of sight start at a single point and is schematically shown in figure 3. .



Figure 2. Photographic image of a series of buildings.

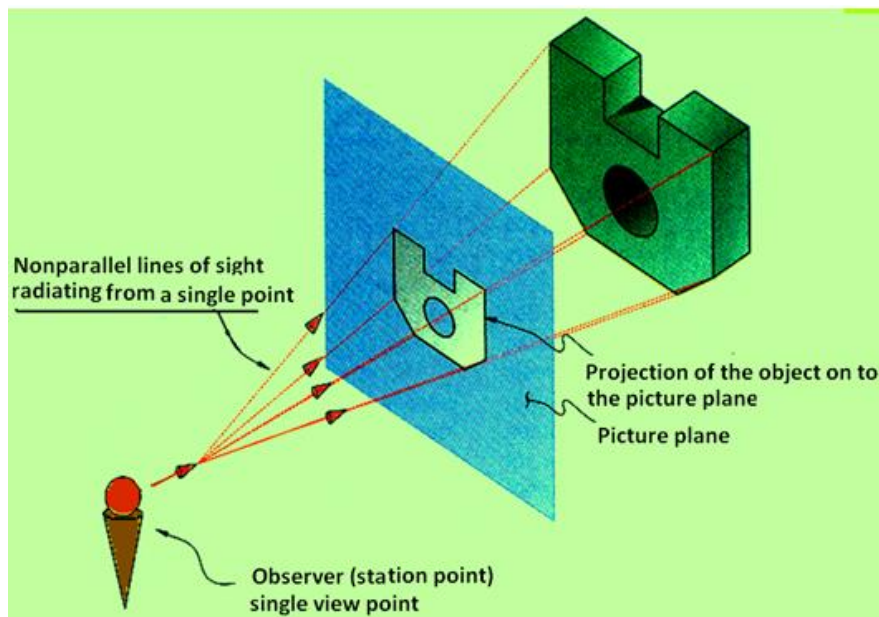


Figure 3. A schematic representation of a Perspective projection

In parallel projection, all lines of sight are parallel and is schematically represented in figure. 4. The observer is assumed to be stationed at infinite distance from the object.

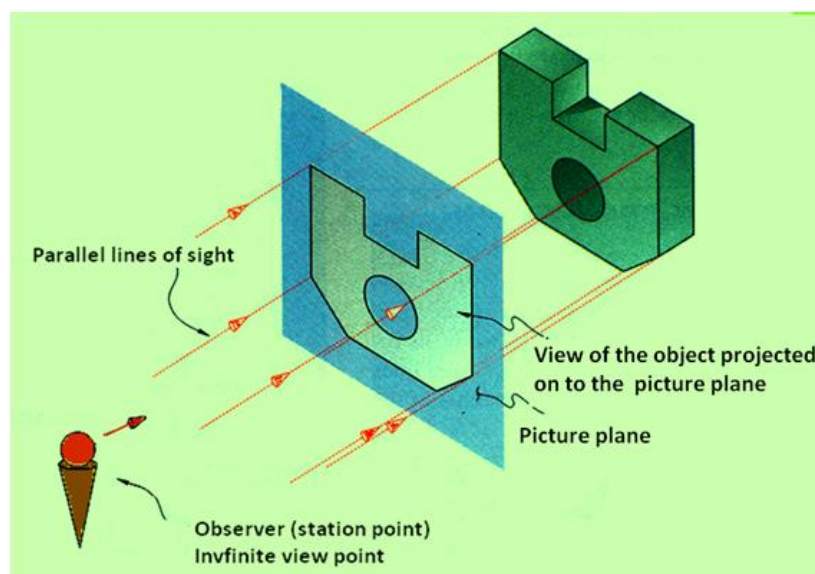


Figure 4. A schematic representation of a Parallel projection

Parallel vs Perspective Projection

Parallel projection

- Distance from the observer to the object is infinite projection lines are parallel – object is positioned at infinity.
- Less realistic but easier to draw.

Perspective projection

- Distance from the observer to the object is finite and the object is viewed from a single point – projectors are not parallel.
- Perspective projections mimic what the human eyes see, however, they are difficult to draw.

Orthographic Projection

Orthographic projection is a parallel projection technique in which the plane of projection is perpendicular to the parallel line of sight. Orthographic projection technique can produce either pictorial drawings that show all three dimensions of an object in one view or multi-views that show only two dimensions of an object in a single view. These views are shown in figure 5.

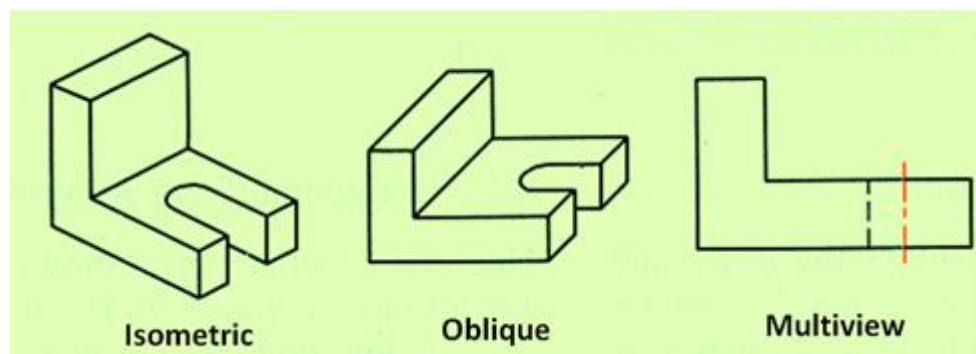


Figure 5. Orthographic projections of a solid showing isometric, oblique and multi-view drawings.

Six Principal views

The plane of projection can be oriented to produce an infinite number of views of an object. However, some views are more important than others. These principal views are the six mutually perpendicular views that are produced by six mutually perpendicular planes of projection and is shown in figure 6. Imagine suspending an object in a glass box with major surfaces of the object positioned so that they are parallel to the sides of the box, six sides of the box become projection planes, showing the six views – front, top, left, right, bottom and rear.

Object is suspended in a glass box producing six principal views: each view is perpendicular to and aligned with the adjacent views.

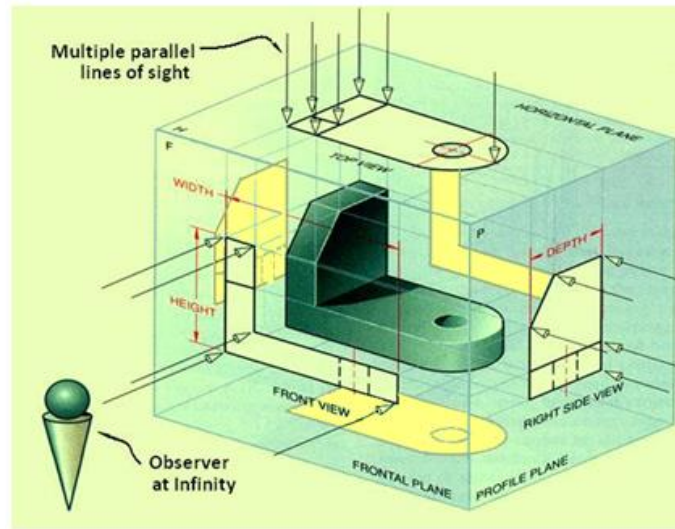


Figure 6. Shows the six perpendicular views of an object

The glass box is now slowly unfolded as shown in figure 7. After complete unfolding of the box on to a single plane, we get the six views of the object in a single plane as shown in figure 8. The top, front and bottom views are all aligned vertically and share the same width dimension whereas the rear, left side, front and right side views are all aligned horizontally and share the same height dimension.

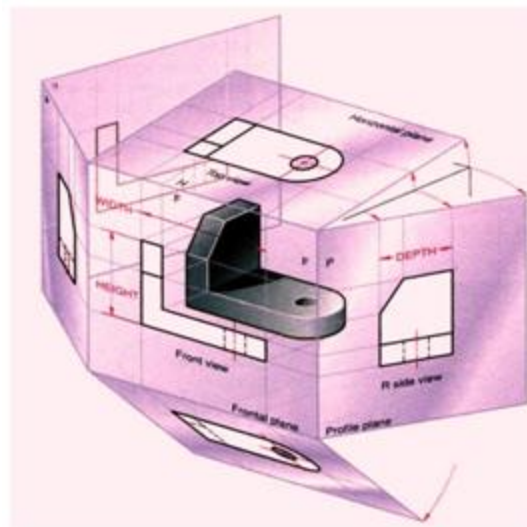


Figure 7. Illustration of the views after the box has been partially unfolded.

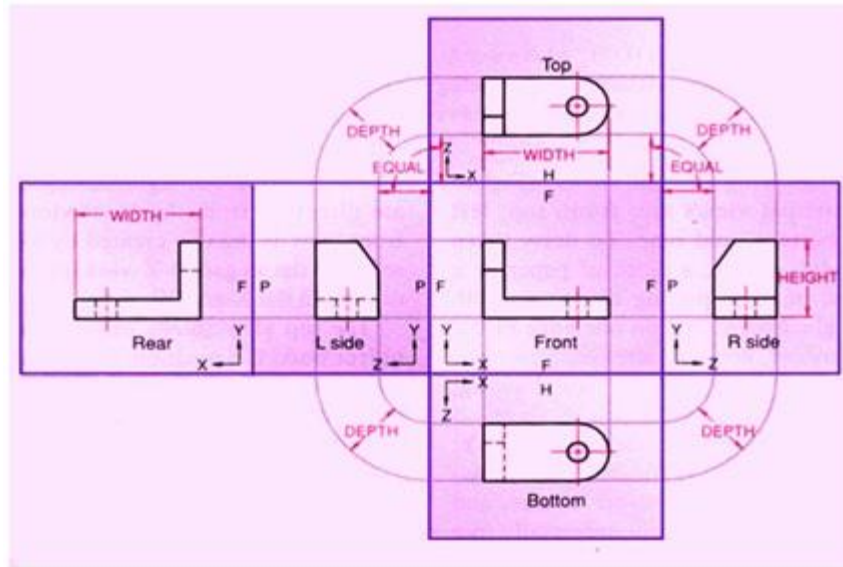


Figure 8 shows the views of the object with their relative positions after the box has been unfolded completely on to a single plane.

Conventional view placement

The three-view multi-view drawing is the standard used in engineering and technology, because many times the other three principal views are mirror images and do not add to the knowledge about the object. Figure 9 shows the standard views used in a three-view drawing i.e., the top, front and the right side views

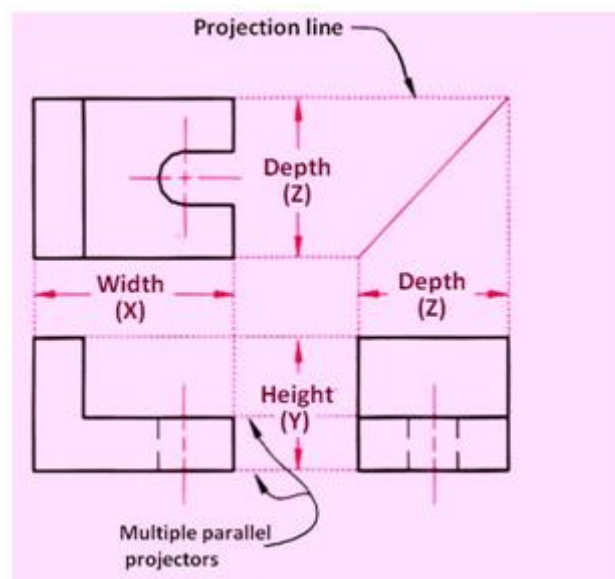


Figure 9 showing the three standard views of a multi-view drawing.

The width dimensions are aligned between the front and top views, using vertical projection lines. The height dimensions are aligned between the front and the profile views, using horizontal projection lines. Because of the relative positioning of the three views, the depth dimension cannot be aligned using projection lines. Instead, the depth dimension is measured in either the top or right side view.

Projection Methods

Universally either the 1st angle projection or the third angle projection methods is followed for obtaining engineering drawings. The principal projection planes and quadrants used to create drawings are shown in figure 10. The object can be considered to be in any of the four quadrant.

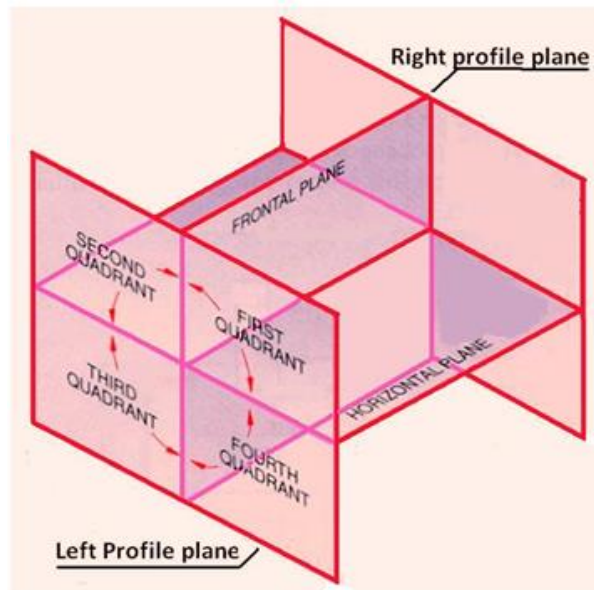


Figure 10. The principal projection planes and quadrants for creation of drawings.

First Angle Projection

In this the object is assumed to be positioned in the first quadrant and is shown in figure 11 the object is assumed to be positioned in between the projection planes and the observer. The views are obtained by projecting the images on the respective planes. Note that the right hand side view is projected on the plane placed at the left of the object. After projecting on to the respective planes, the bottom plane and left plane is unfolded on to the front view plane. i.e. the left plane is unfolded towards the left side to obtain the Right hand side view on the left side of the Front view and aligned with the Front view. The bottom plane is unfolded towards the bottom to obtain the Top view below the Front view and aligned with the Front View.

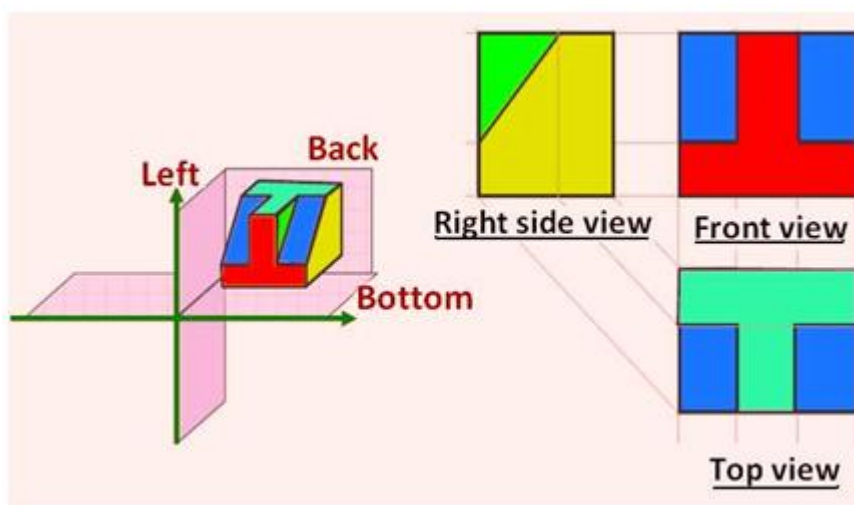


Figure 11. Illustrating the views obtained using first angle projection technique.

Third Angle Projection

In the third angle projection method, the object is assumed to be in the third quadrant. i.e. the object behind vertical plane and below the horizontal plane. In this projection technique, placing the object in the third quadrant puts the projection planes between the viewer and the object and is shown in figure 12.

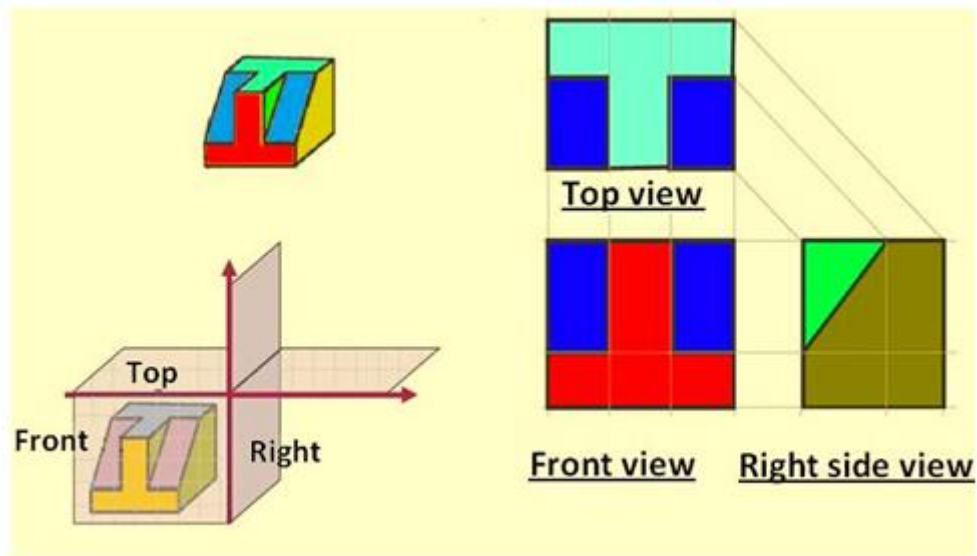


Figure 12. Illustrating the views obtained using first angle projection technique

Figure 13 illustrates the difference between the 1st angle and 3rd angle projection techniques. A summary of the difference between 1st and 3rd angle projections is shown if Table 1.

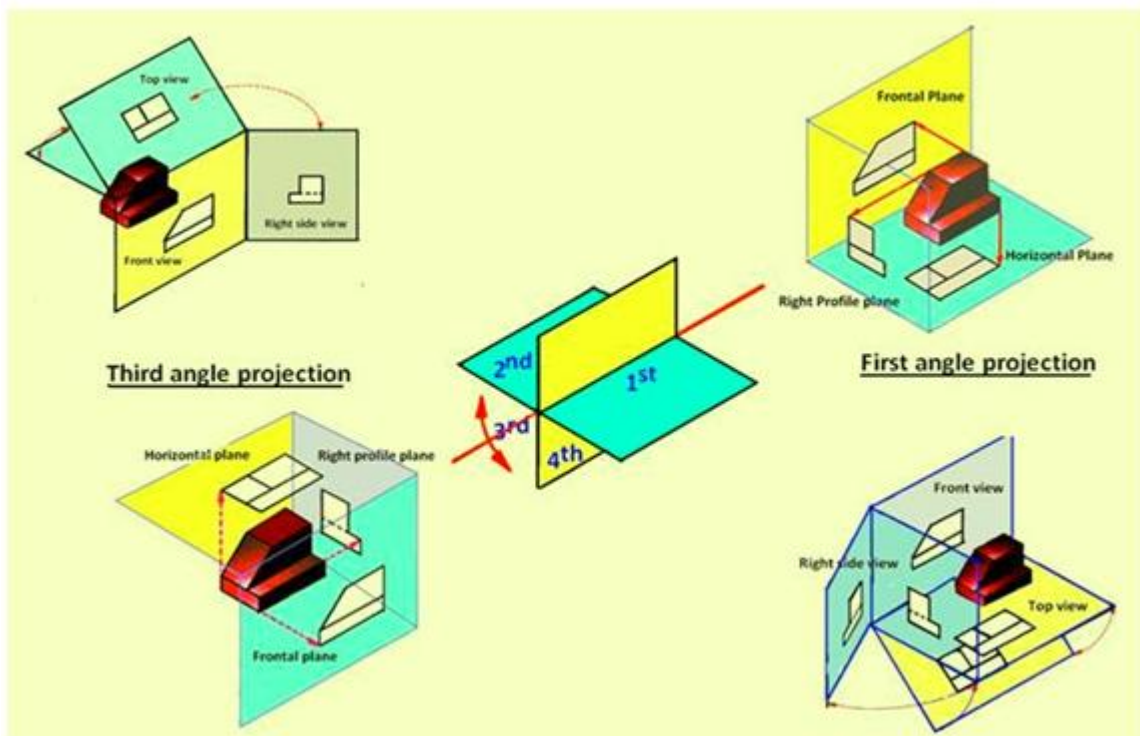


Figure 13 Differentiating between the 1st angle and 3rd angle projection techniques.

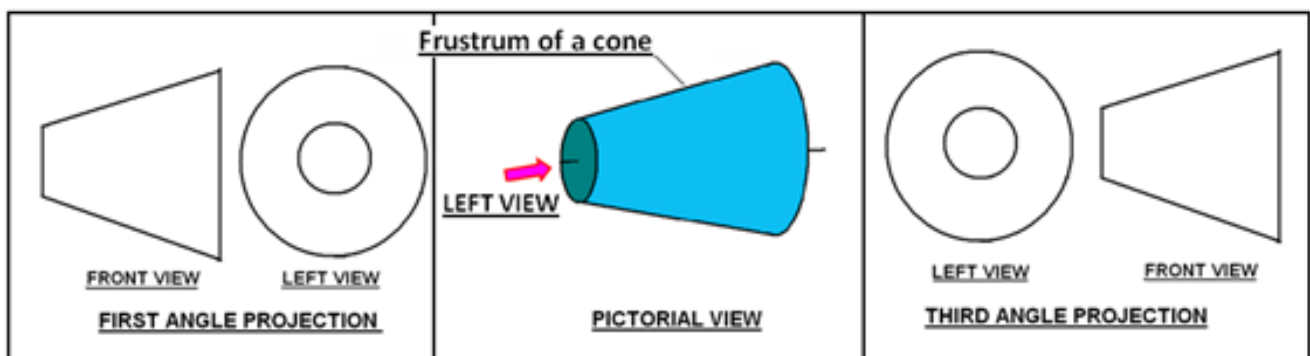
Table 1. Difference between first- and third-angle projections

First angle projection	Third-angle projection
Object is kept in the first quadrant.	Object is assumed to be kept in the third quadrant.
Object lies between observer and the plane of projection.	Plane of projection lies between the observer and the object.
The plane of projection is assumed to be non-transparent.	The plane of projection is assumed to be transparent.
Front (elevation) view is drawn above the XY line	Front (elevation) view is drawn below the XY line
Top (plan) view is drawn below the XY line	Top (plan) view is drawn above the XY line
Left view is projected on the right plane and vice versa	Left view is projected on the left plane itself.
Followed in India, European countries	Followed in USA

Either first angle projection or third angle projection are used for engineering drawing. Second angle projection and fourth angle projections are not used since the drawing becomes complicated. This is being explained with illustrations in the lecture on Projections of points.

Symbol of projection

The type of projection obtained should be indicated symbolically in the space provided for the purpose in the title box of the drawing sheet. The symbol recommended by BIS is to draw the two sides of a frustum of a cone placed with its axis horizontal the left view is drawn.



Orthographic Projections

Lines are used to construct a drawing. Various type of lines are used to construct meaningful drawings. Each line in a drawing is used to convey some specific information. The types of lines generally used in engineering drawing is shown in Table-1.

Table -1. Types of lines generally used in drawings

a) Visible lines	
b) Hidden lines	
c) Centre lines	
d) Dimension lines	
e) construction lines	

All visible edges are to be represented by visible lines. This includes the boundary of the object and intersection between two planes. All hidden edges and features should be represented by dashed lines. Figure 1 shows the orthographic front view (line of sight in the direction of arrow) of an object. The external boundary of the object is a rectangle and is shown by visible lines. In Figure-1(a), the step part of the object is hidden and hence shown as dashed lines while for the position of the object shown in figure-1(b), the step part is directly visible and hence shown by the two solid lines.

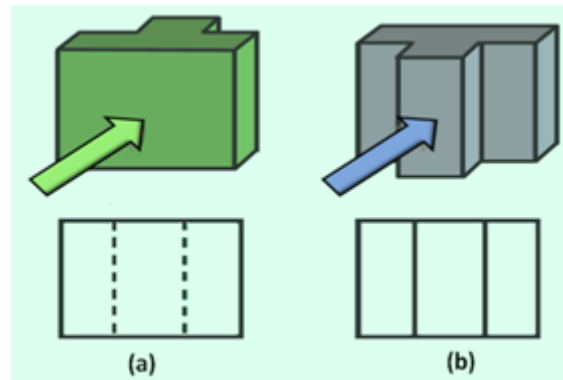


Figure 1 shows the pictorial view and front view of the object when the middle stepped region is

(a) Hidden and (b) visible.

Figure 2 shows the front view (view along the direction indicated by the arrow) of a solid and hollow cylindrical object. The front view of the solid cylinder is seen as a rectangle (figure 2(a)). For the hollow cylinder in addition to the rectangle representing the boundary of the object, two dashed lines are shown to present the boundary of the hole, which is a hidden feature in the object.

Figure 3 shows the Front view of three objects. Figure 3(a) is the view of one part of a hollow cylinder which has been split in to two equal parts. The wall thickness can be represented by the two visible lines. Figure 3(b) is one part of solid cylinder which has been sectioned in to two equal part. Whereas figure 3(c) is one part of a solid cylindrical part which has been split in to two unequal parts. The edge formed by the intersection of two surfaces are represented by solid lines. In case of cylindrical objects or when holes are present in a component, the centre of the holes or centre lines of cylinder will have to be represented in the drawing by means of centre lines as shown in figure 4. Figure 5 shows the FV, TV, and RHSV of an object showing visible edges, hidden edges (or holes), and centre lines.

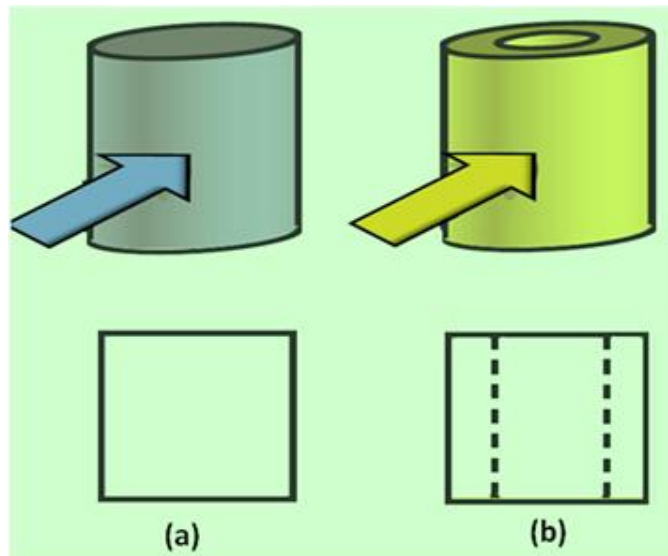


Figure 2 shows the pictorial view and front view of (a) a hollow cylindrical object and (b) solid cylindrical object.

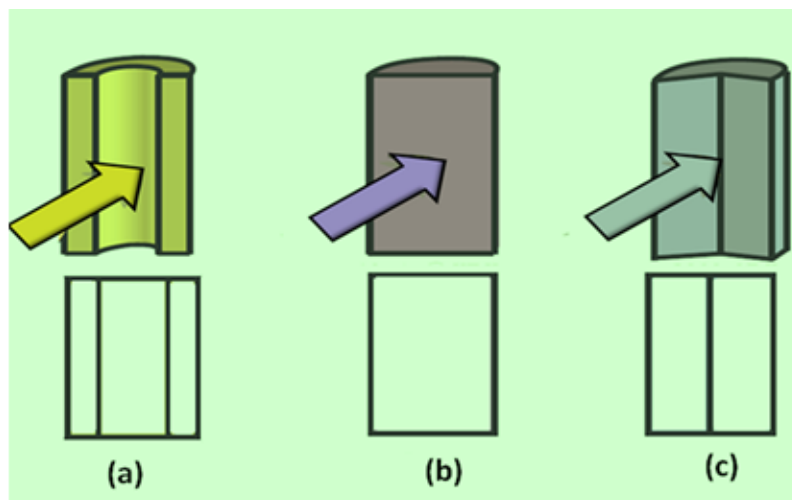


Figure 3 shows the pictorial view and front view of sectioned part of (a) a hollow cylindrical object (b) solid cylindrical object and (c) solid cylinder split in to two unequal parts.

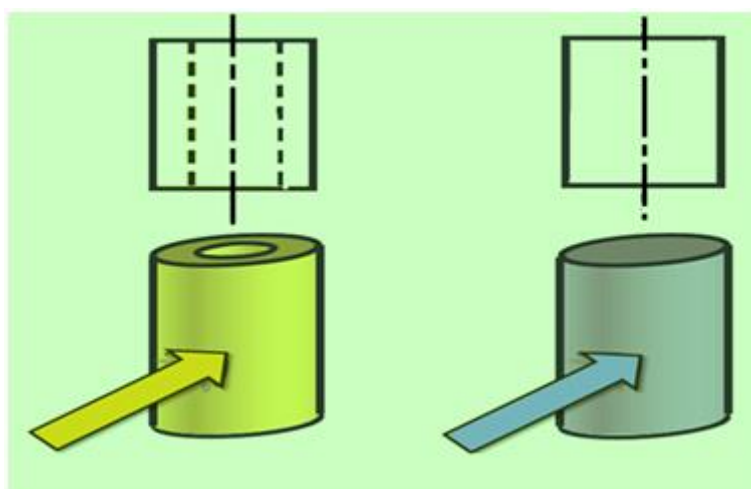


Figure 4 shows the centre lines for cylindrical objects

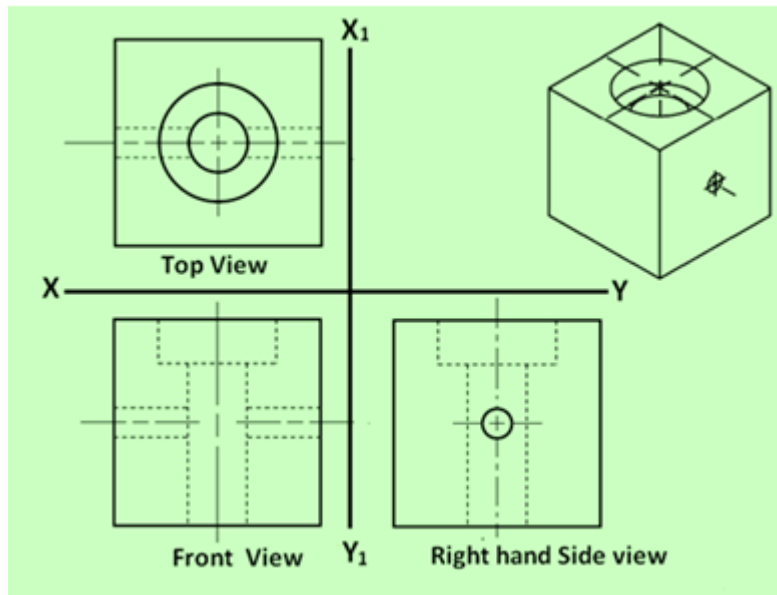


Figure 5. Showing TV, FV and RHSV of an object showing the three types of lines mentioned above. The pictorial view of the object is shown at the top right hand side.

Conventions used for lines

In orthographic projections, many times different types of lines may fall at the same regions. In such cases, the following rules for precedence of lines are to be followed:

- Visible lines take precedence over all other lines
- Hidden lines take precedence over center lines
- Center lines have lowest precedence

When a visible line and a hidden line are to be drawn at the same area, it will be shown by the visible line only and no hidden line will be shown. Similarly, in case of hidden line and centre line, only hidden line will be shown. In such case, the centre line will be shown only if it is extending beyond the length of the hidden line.

Intersecting Lines in Orthographic Projections

The conventions used when different lines intersect is shown in figure - 6(a) & (b).

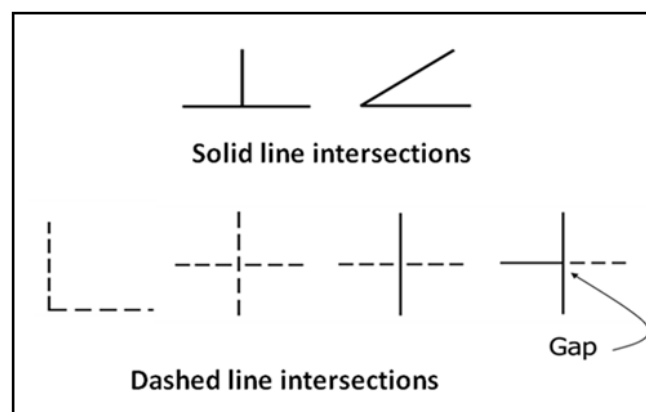


Figure 6(a): The conventions practiced for intersection lines.

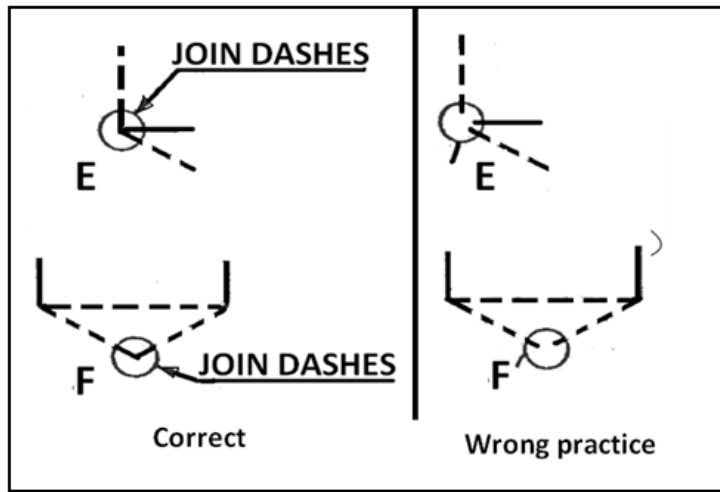
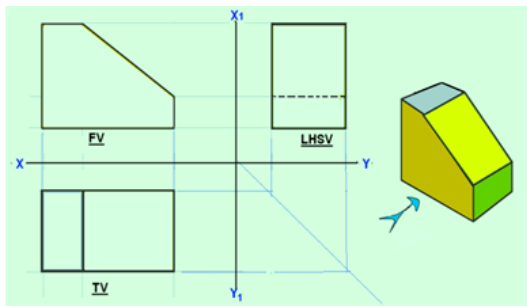


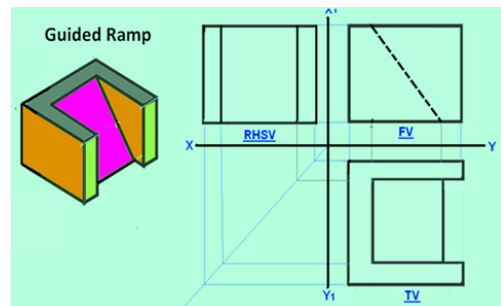
Figure 6(b): The conventions practiced for intersection lines.

Some orthographic projections of solids showing the different lines and their precedence are shown as examples below. The 3-D view of the respective objects are also shown in the figures with the direction of arrow representing the line of sight in the front view.

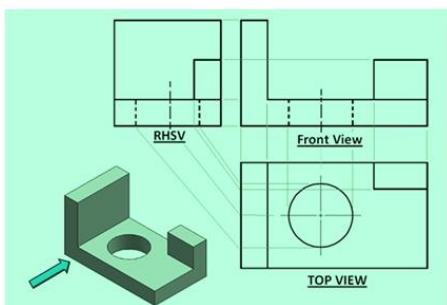
A few examples of the projections showing the conventions in drawing are presented below.



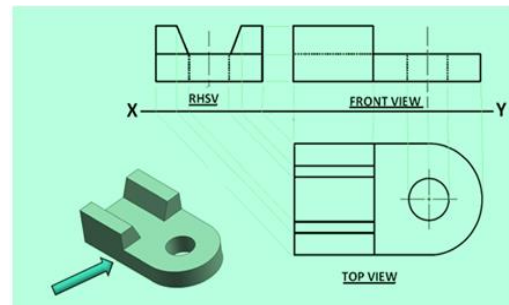
Example 1



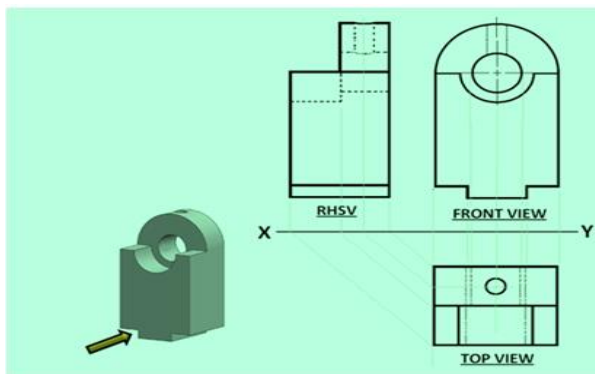
Example 2



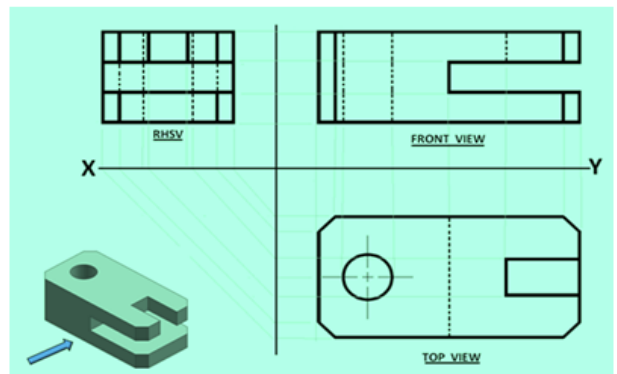
Example 3



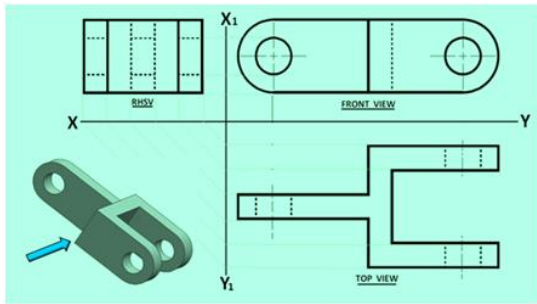
Example 4



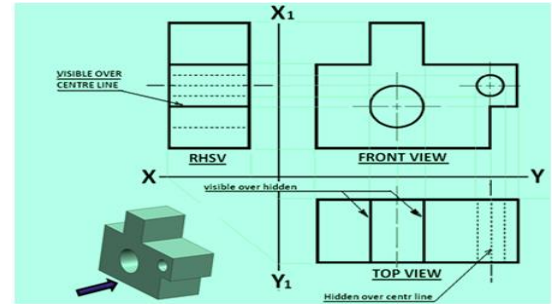
Example 5



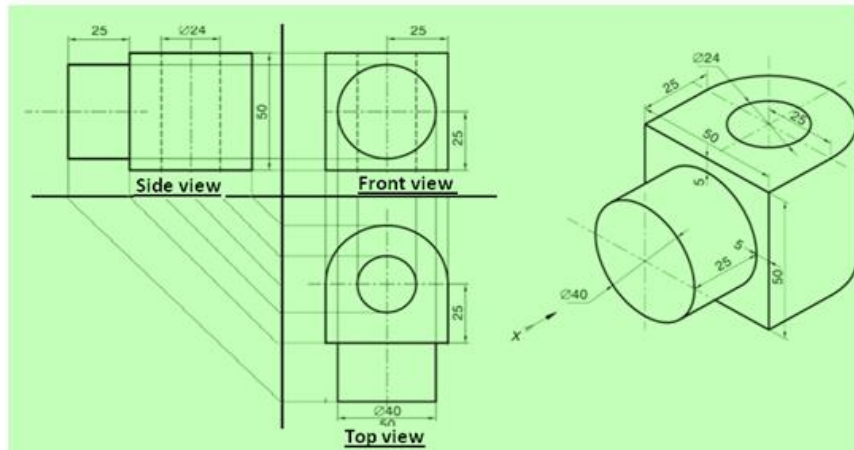
Example 6



Example 7



Example 8 (application of Precedence rule)



Example 9 (Objects with circular features: holes, flanges, etc)

Projection of Points

A POINT

The position of a point in engineering drawing is defined with respect to its distance from the three principle planes i.e., with respect to the VP, HP, & PP.

The point is assumed to be in the respective quadrant shown in figure 1(a). The point at which the line of sight (line of sight is normal to the respective plane of projection) intersects the three planes are obtained. The horizontal plane and the side planes are rotated so such that they lie on the plane containing the vertical plane. The direction of rotation of the horizontal plane is shown in figure 1 (b).

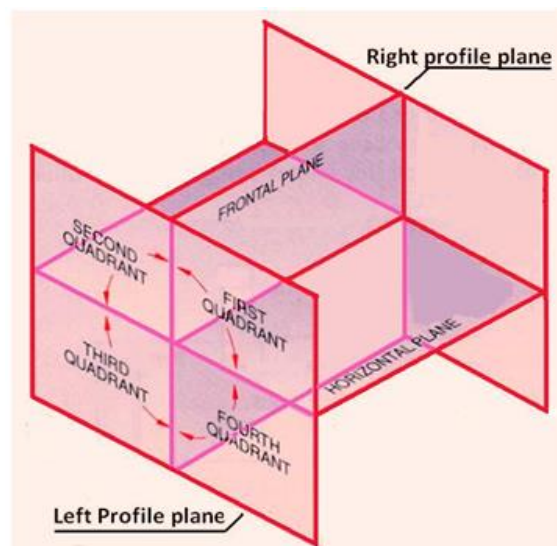


Figure 1(a). The relative positions of projection planes and the quadrants

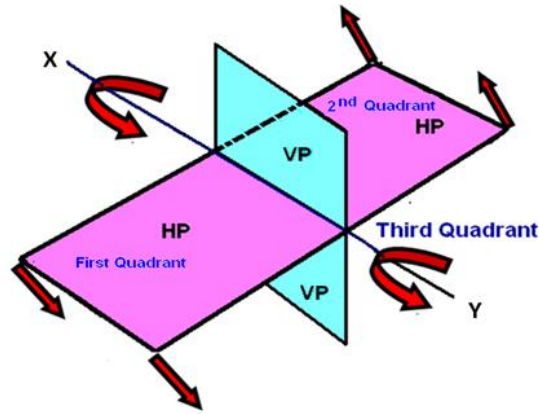


Figure 1(b). The direction of rotation of the Horizontal plane.

Conventions used while drawing the projections of points

- With respect to the 1st angle projection of point “P” shown in figure 2,
- Top views are represented by only small letters eg. p .
- Their front views are conventionally represented by small letters with dashes eg. p’
- Profile or side views are represented by small letters with double dashes eg. p’’
- Projectors are shown as thin lines.
- The line of intersection of HP and VP is denoted as X-Y.
- The line of intersection of VP and PP is denoted as X1-Y1

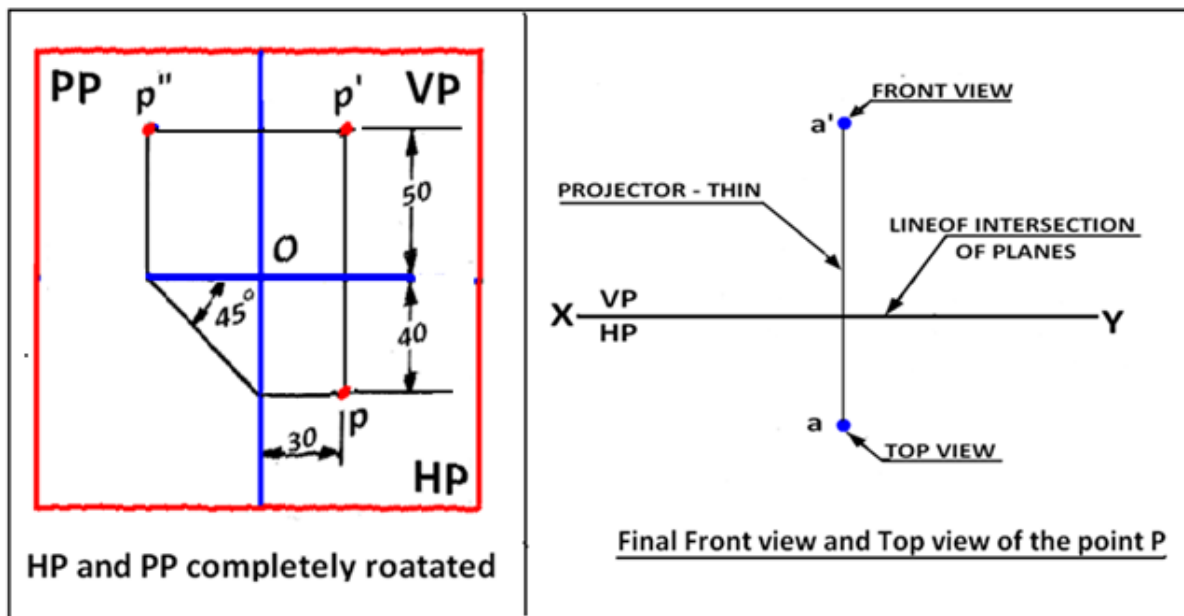


Figure 2. Showing the three planes and the projection of the point P after the planes have been rotated on to the vertical plane.

Point in the First quadrant

Figure 3 shown the projections of a point P which is 40 mm in front of VP, 50 mm above HP, 30 mm in front of left profile plane (PP)

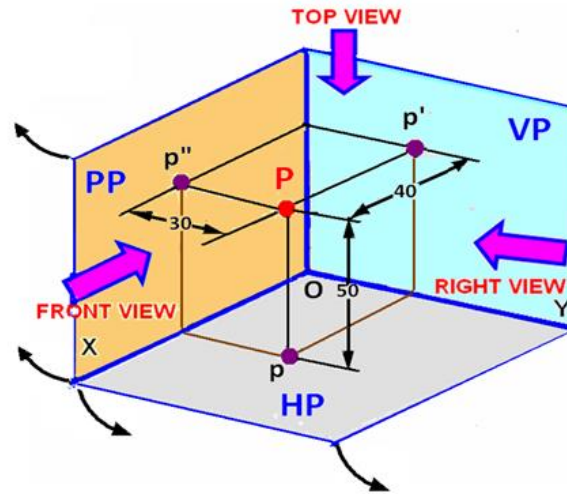


Figure 3. Projection of the point “P” on to the three projection planes before the planes are rotated.

Figure 4 shows the planes and the position of the points when the planes are partially rotated. The arrows indicate the direction of rotation of the planes. The three views after complete rotation of the planes is shown in figure 2.

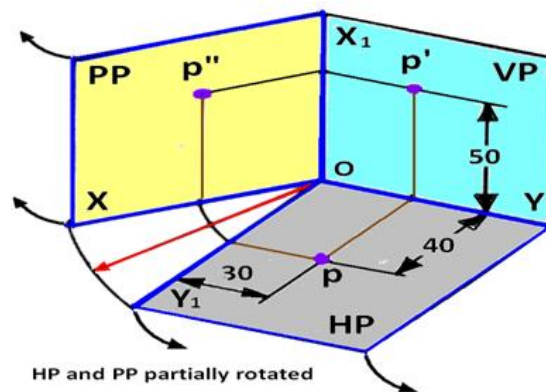


Figure 4. Projection of the point “P” on to the three projection planes after the planes are partially rotated.

- The procedure of drawing the three views of the point “P” is shown in figure-4.
- Draw a thin horizontal line, XY, to represent the line of intersection of HP and VP.
- Draw X1Y1 line to represent the line of intersection of VP and PP.
- Draw the Top View (p).
- Draw the projector line
- Draw the Front View (p') .
- To project the right view on the left PP, draw a horizontal projector through p to intersect the 45 degree line at m. Through m draw a vertical projector to intersect the horizontal projector drawn through p' at p''.
- p'' is the right view of point P

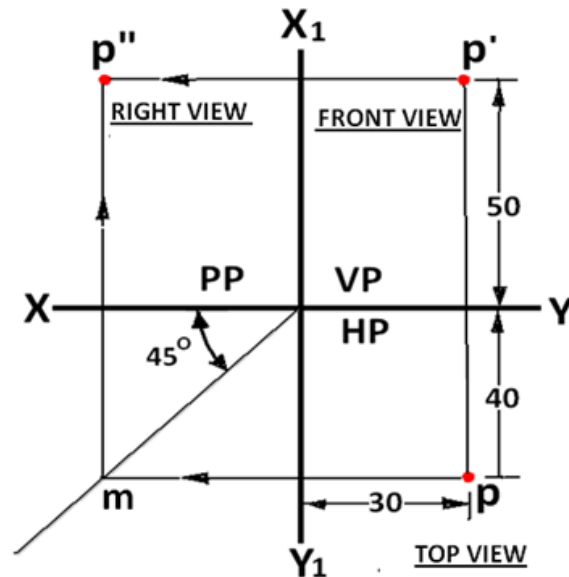


Figure 5 First angle multi-view drawing of the point “P”

Point in the Second quadrant

Point P is 30 mm above HP, 50 mm behind VP and 45 mm in front of left PP. Since point P is located behind VP, the VP is assumed transparent. The position of the point w.r.t the three planes are shown in Figure 1. The direction of viewing are shown by arrows. After projecting the point on to the three planes, the HP and PP are rotated such that they lie along the VP. The direction of rotation of the HP and PP is shown in figure 2. As shown in figure 3, after rotation of the PP and HP, it is found that the VP and HP is overlapping. The Multiview drawing for the point P lying in the second quadrant is shown in figure 4. Though for the projection of a single point, this may not be a problem, the Multiview drawing of solids, where a number of lines are to be drawn, will be very complicated. Hence second angle projection technic is not followed anywhere for engineering drawing.

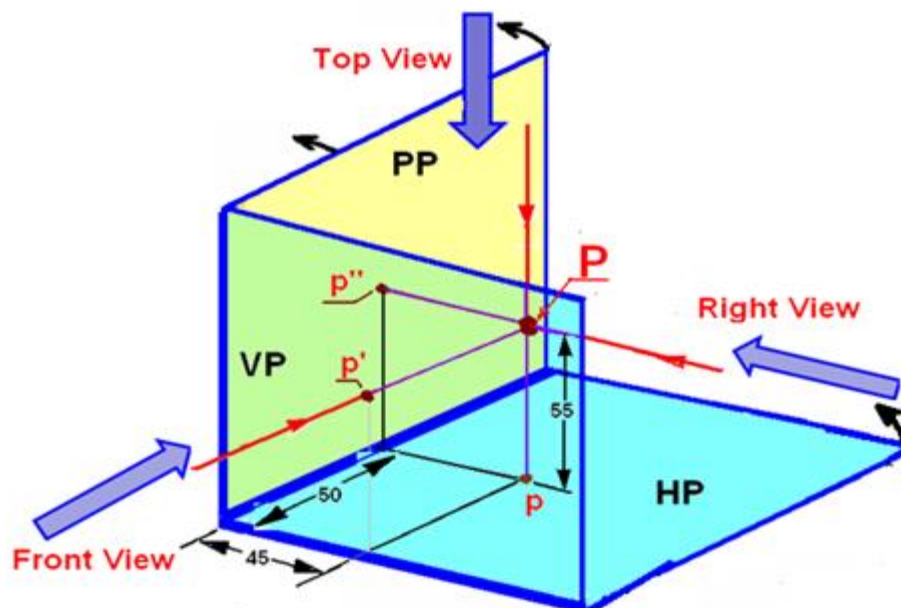


Figure 1. The projection of point P on to the three projection planes.

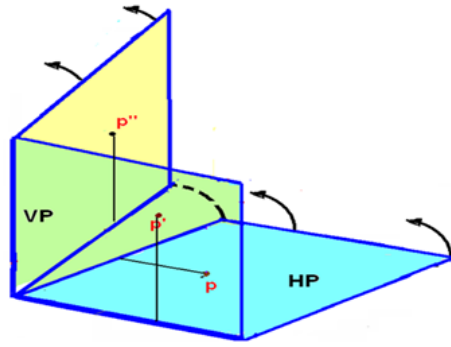


Figure 2. The direction of rotation of HP.

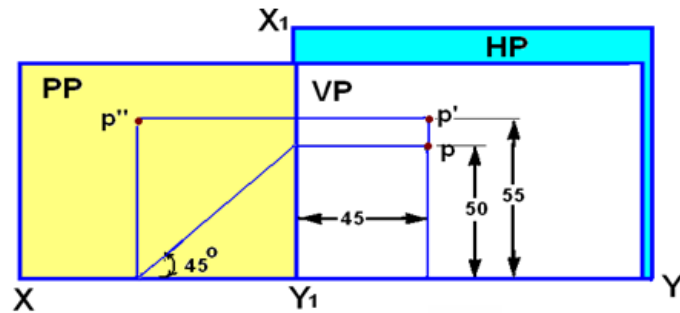


Figure 3. The projection of point P after complete rotation of the HP and PP.

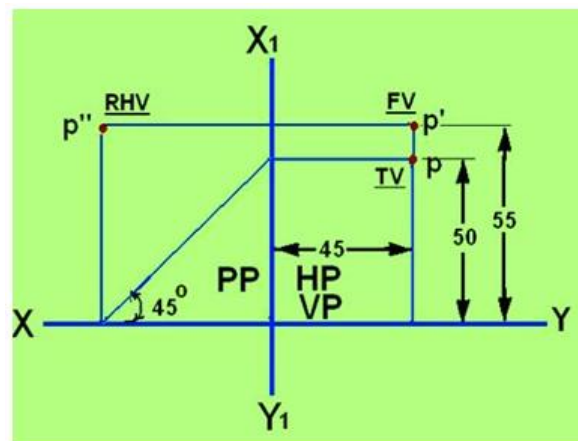


Figure 4. The Multiview drawing of the point P lying in the second quadrant.

Point in the Third quadrant

Projection of a point P in the third quadrant where P is 40 mm behind VP, 50 mm below HP and 30 mm behind the right PP is shown in figure 5.

Since the three planes of projections lie in between the observer and the point P, they are assumed as transparent planes. After the point P is projected on to the three planes, the HP and VP are rotated along the direction shown in figure 6, such that the HP and PP is in plane with the VP. The orthographic projection of the point P lying in the third quadrant is shown in figure 7.

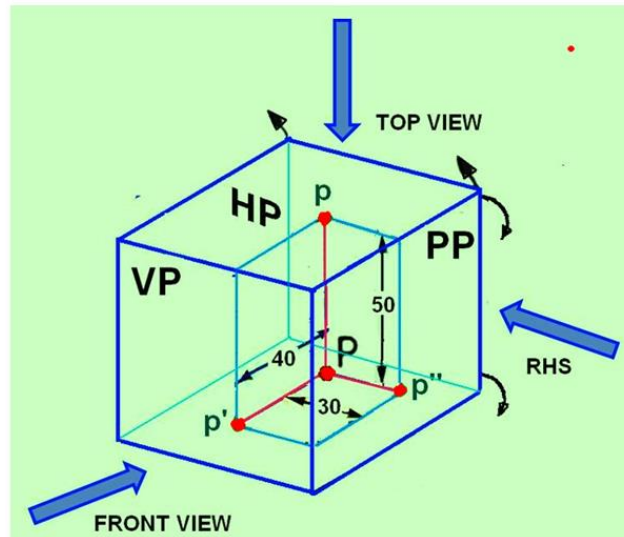


Figure 5. Projection of a point P placed in the third quadrant

In the third angle projection, the Top view is always above the front view and the Right side view will be towards the right of the Front view.

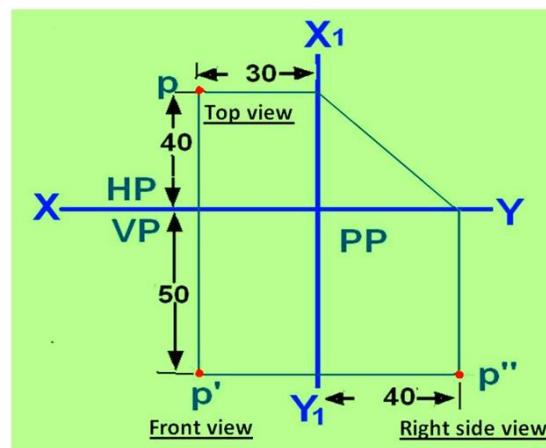


Figure 7. Multi-view drawing of the point lying in the third quadrant.

In the third angle projection, the Top view is always above the front view and the Right side view will be towards the right of the Front view.

Point in the Fourth quadrant

If A point is lying in the fourth quadrant, the point will be below the HP and in front of the VP. The point is projected on to the respective projection planes. After rotation of the HP and PP on to the VP, it will be observed that the HP and VP are overlapping, similar to the second angle projection. The multi-view drawing of objects in such case would be very confusing and hence fourth angle projection technique is not followed by engineers.

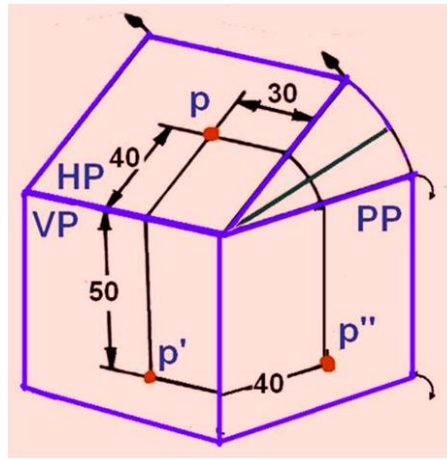


Figure 6. Shows the sense of direction of rotation of PP and HP.

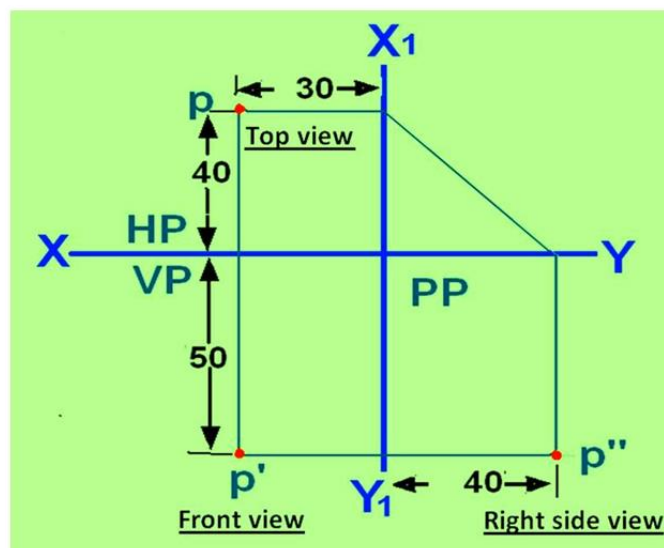


Figure 7. Multi-view drawing of the point lying in the third quadrant.

In the third angle projection, the Top view is always above the front view and the Right side view will be towards the right of the Front view.

Point in the Fourth quadrant

If a point is lying in the fourth quadrant, the point will be below the HP and in front of the VP. The point is projected on to the respective projection planes. After rotation of the HP and PP on to the VP, it will be observed that the HP and VP are overlapping, similar to the second angle projection. The multi-view drawing of objects in such case would be very confusing and hence fourth angle projection technique is not followed by engineers.

Shear stresses:

Let us consider now the situation, where the cross – sectional area of a block of material is subject to a distribution of forces which are parallel, rather than normal, to the area concerned. Such forces are associated with a shearing of the material, and are referred to as shear forces. The resulting force interests are known as shear stresses.

Projections of lines

Straight line

A line is a geometric primitive that has length and direction, but no thickness. Straight line is the Locus of a point, which moves linearly. Straight line is also the shortest distance between any two given points.

The location of a line in projection quadrants is described by specifying the distances of its end points from the VP, HP and PP. A line may be:

- Parallel to both the planes.
- Parallel to one plane and perpendicular to the other.
- Parallel to one plane and inclined to the other.
- Inclined to both the planes.
- Projection of a line

The projection of a line can be obtained by projecting its end points on planes of projections and then connecting the points of projections. The projected length and inclination of a line, can be different compared to its true length and inclination.

Case 1. Line parallel to a plane

When a line is parallel to a plane, the projection of the line on to that plane will be its true length. The projection of line AB lying parallel to the Vertical plane (VP) is shown in figure 1 as a'b'.

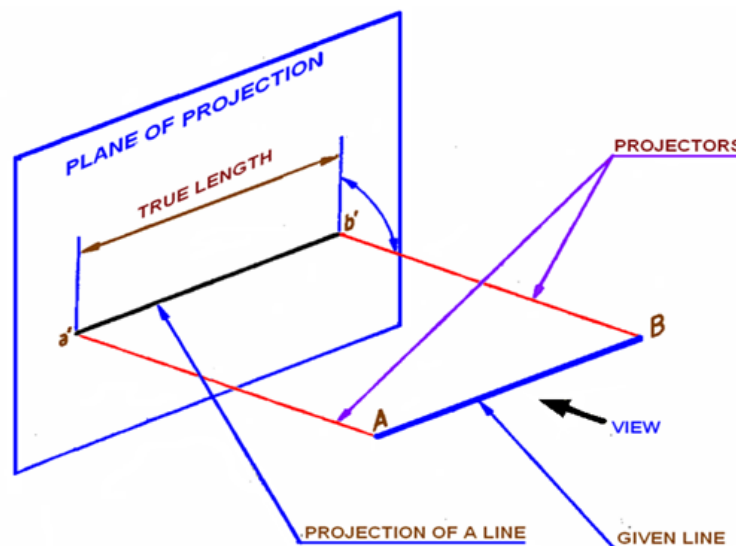


Figure 1. Projection of line on VP. Line AB is parallel to VP.

Case 2. Line inclined to a plane

When a line is parallel to one plane and inclined to the other, the projection of the line on the plane to which it is parallel will show its true length. The projected length on the plane to which it is inclined will always be shorter than the true length. In figure 2, the line AB is parallel to VP and is inclined to HP. The angle of inclination of AB with HP is being θ degrees. Projection of line AB on VP is a'b' and is the true length of

AB. The projection of line AB on HP is indicated as line ab. Length ab is shorter than the true length AB of the line.

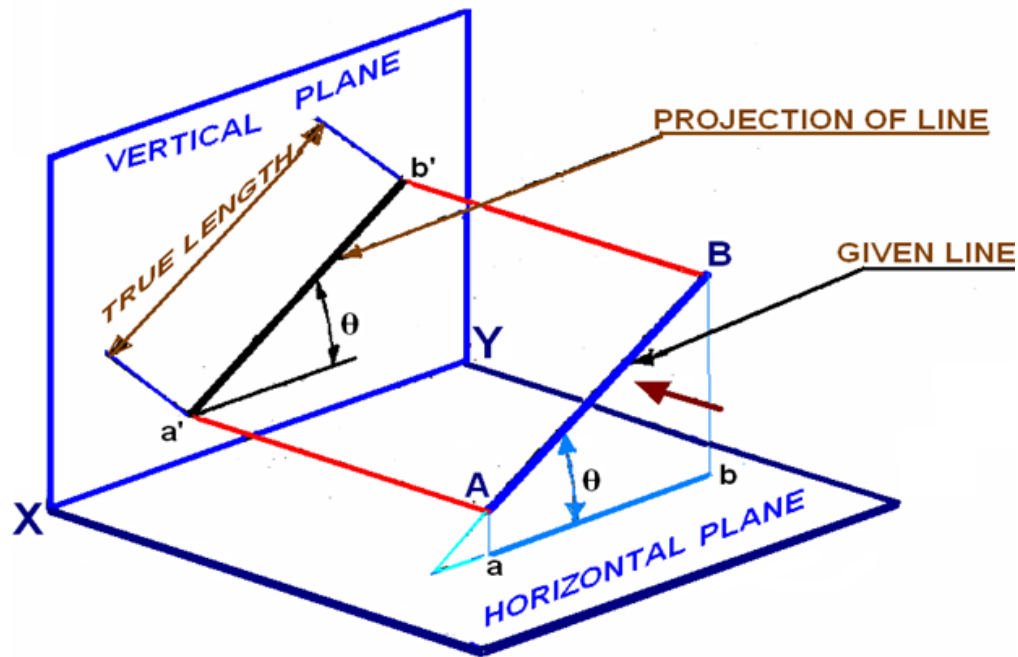


Figure 2. Projection of line AB parallel to VO and inclined to HP.

Case 3. Projection of a line parallel to both HP and VP

A line AB having length 80 mm is parallel to both HP and VP. The line is 70 mm above HP, 60 mm in front of VP. End B is 30 mm in front of right PP. Draw the projection of line AB, assume the line in the first quadrant.

The projection points of AB on the vertical plane VP, horizontal plane HP and Right Profile plane PP is shown in figure 3(a). Since the line is parallel to both HP and VP, both the front view $a'b'$ and the top view ab are in true lengths. Since the line is perpendicular to the right PP, the left side view of the line will be a point $a''(b'')$. After projection on to the projection planes, the planes are rotated such that all the three projection planes lie in the same planes. The multi-view drawing of line AB is shown in Figure 3(b).

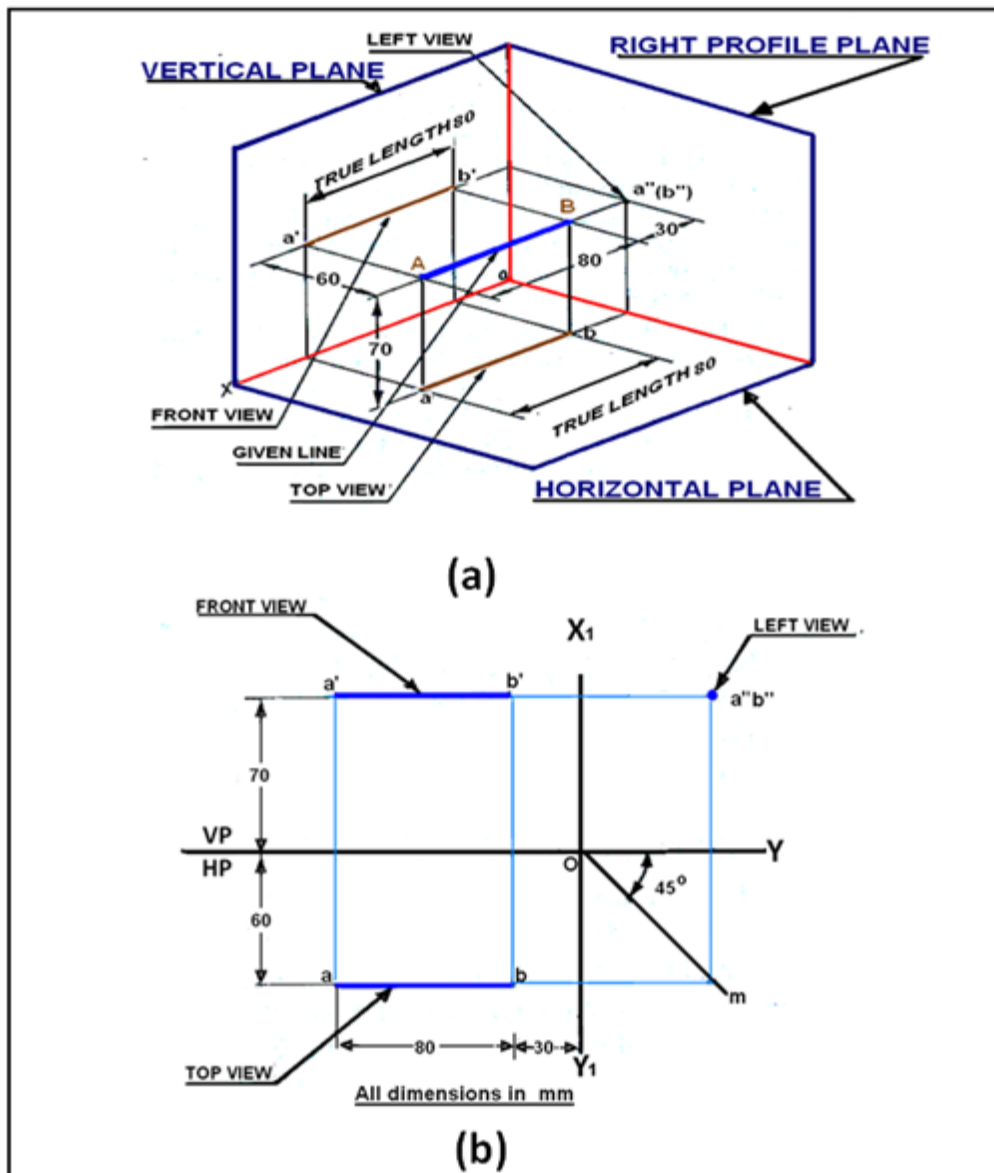


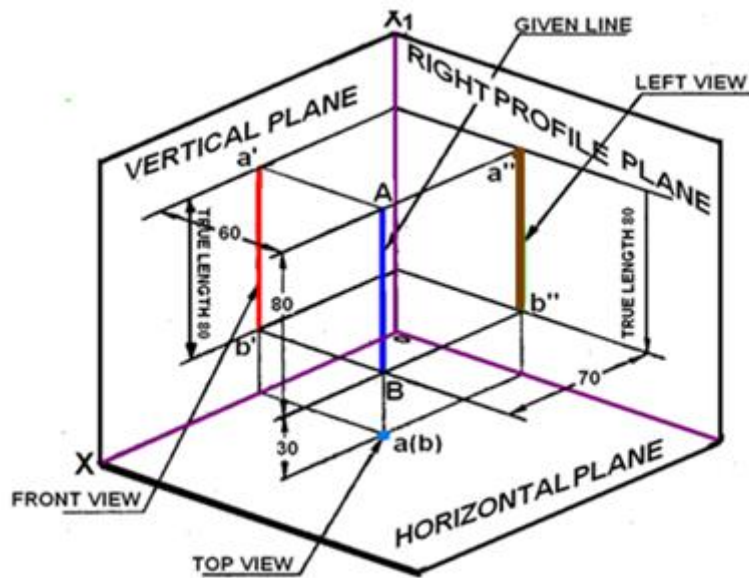
Figure 3. Projection of line parallel to both HP and VP.

Case 4. Line perpendicular to HP & parallel to VP

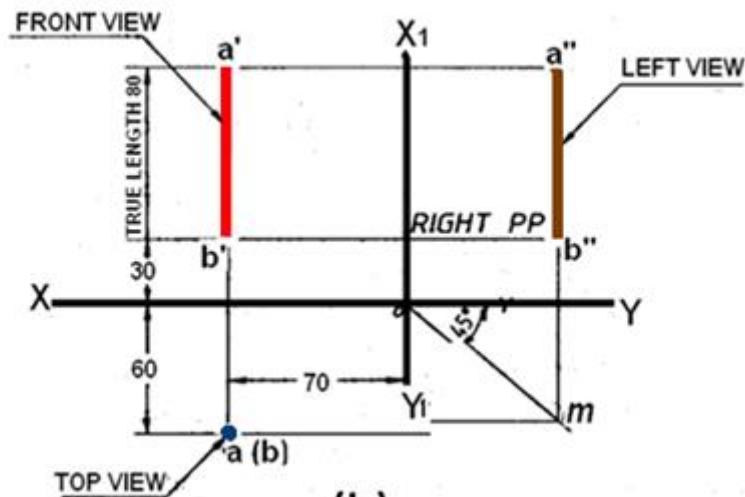
A line AB of length 80 mm is parallel to VP and perpendicular to HP. The line is 80 mm in front of VP and 80 mm in front of right PP. The lower end of the line is 30 mm above HP. The projections of line AB shown in figure 4 can be obtained by the following method.

Draw a line XY which is the intersection between VP and HP. Draw the front view $a'b' = 80$ mm perpendicular to the XY line, with the lower end b' lying 30 mm above the XY line. Project the top view of the line which will be a point $a(b)$ at a distance of 60 mm below XY line. Since the line is 70 mm in front of the right PP draw the X_1Y_1 line at a distance of 70 mm on the right- side of the front view.

Through O the point of intersection of XY and X_1Y_1 , lines draw a 45° line. Draw the horizontal projector through $a(b)$ to cut the 45 degree line at m. Draw the horizontal projectors through a' and b' to intersect the vertical projector drawn through m at a'' and b'' . $a''b''$ is the left view of the line AB.



(a)



(b)

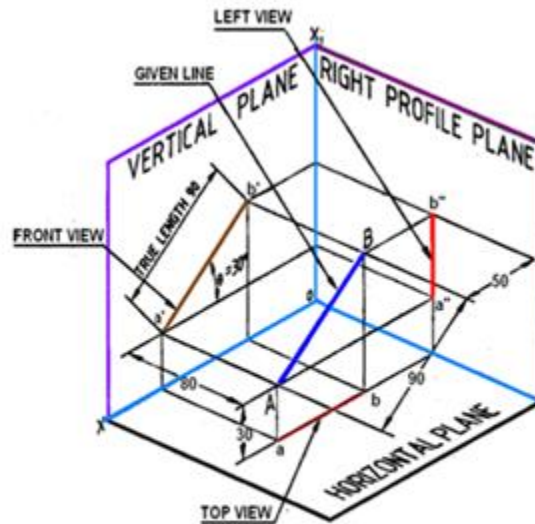
Figure 4. Projections of a line AB perpendicular to HP and parallel to VP.

Line parallel to one plane and inclined to the other

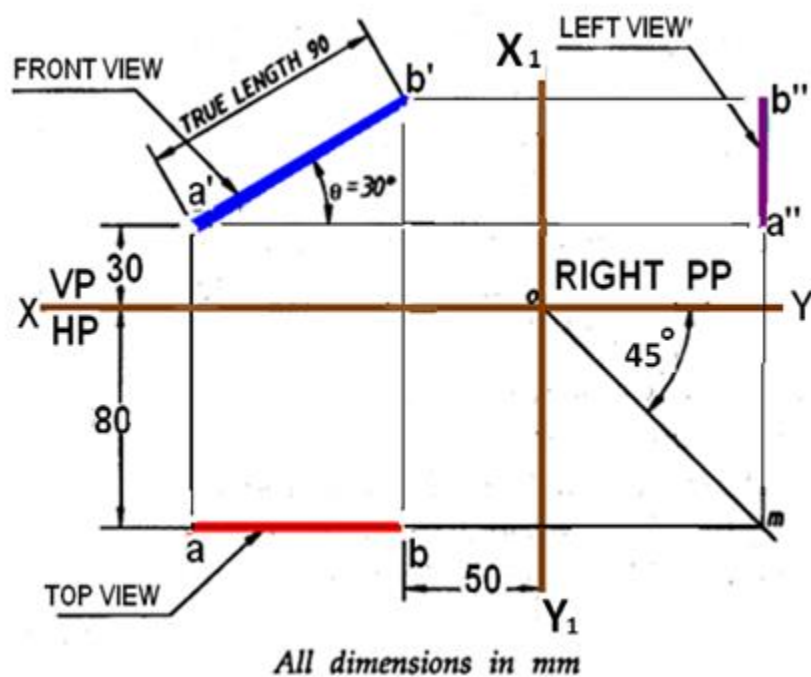
Case 5. Line parallel to VP and inclined to HP

A line AB, 90 mm long is inclined at 30° to HP and parallel to VP. The line is 80 mm in front of VP. The lower end A is 30 mm above HP. The upper end B is 50 mm in front of the right PP. The projections of line AB shown in figure 5 can be obtained in the following manner. Mark a' , the front view of the end A, 30 mm above HP. Draw the front view $a'b' = 90$ mm inclined at 30° to XY line.

Project the top view ab parallel to XY line. The top view is 80 mm in front of VP. Draw the X_1Y_1 line at a distance of 50 mm from b' . Draw a 45° line through O. Draw the horizontal projector through the top view ab to cut the 45° line at m . Draw a vertical projector through m . Draw the horizontal projectors through a' and b' to intersect the vertical projector drawn through m at a'' and b'' . Connect $a''b''$ which is the left side view.



(a)



(b)

Figure 5. Projections of line AB parallel to VP and inclined to HP.

Case 6. Line inclined to HP and VP

When a line is inclined to both HP and VP, the apparent inclination of the line to both the projection planes will be different from the actual inclinations. Similarly the projected length of the lines on to the planes will not be the same as the true length of the line. The following notation will be used for the inclinations and length of the lines for this entire lecture series:

Actual inclinations are θ degrees to HP and ϕ degrees to VP Apparent Inclinations are α and β to HP and VP respectively.

The Apparent Lengths of line AB are ab and $a'b'$ in the top view and front view respectively.

Example: Draw the projections of a line AB inclined to both HP and VP, whose true length and true inclinations and locations of one of the end points, say A are given.

The projections of the line AB are illustrated in figure 1. Since the line AB is inclined at θ to HP and ϕ to VP – its top view ab and the front view $a'b'$ are not in true lengths and they are also not inclined at angles θ to HP and ϕ to VP in the Front view and top view respectively. Figure 2 illustrates the projections of the line AB when the line is rotated about A and made parallel to VP and HP respectively. A clear understanding of these can be understood if the procedure followed in the subsequent sub-sections are followed:

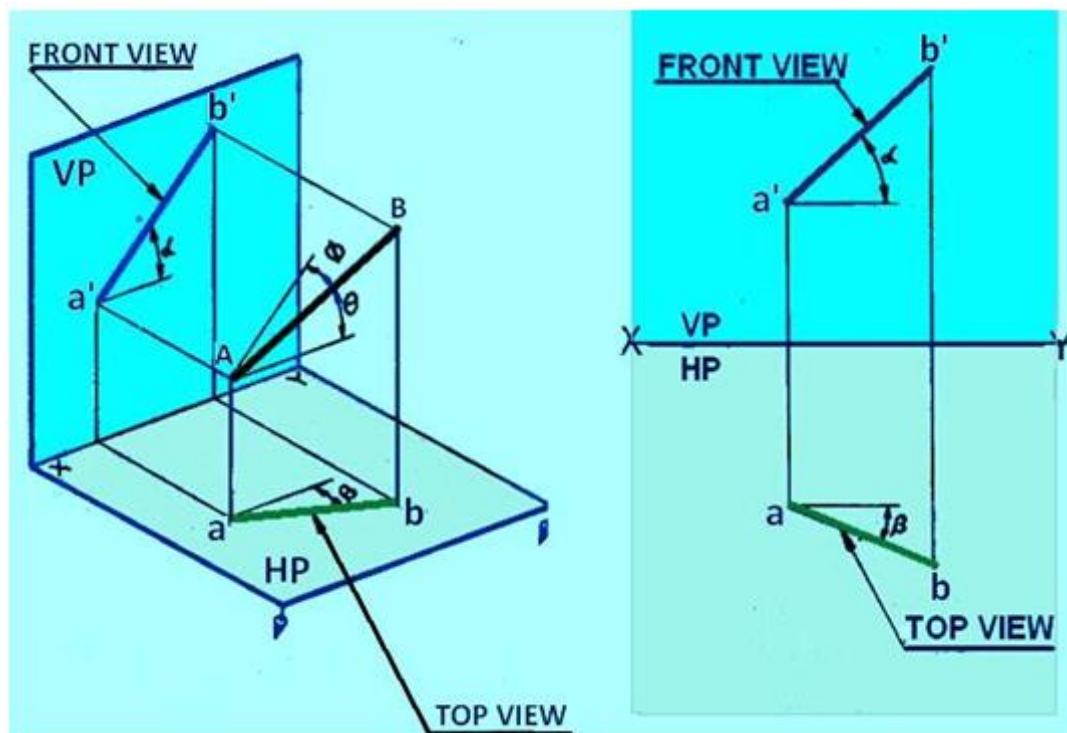


Figure 1: The projections of a line inclined to both HP and VP

Step 1: Rotate the line AB to make it parallel to VP.

Rotate the line AB about the end A, keeping θ , the inclination of AB with HP constant till it becomes parallel to VP. This rotation of the line will bring the end B to the new position B1. AB1 is the new position of the line AB when it is inclined at ϕ to HP and parallel to VP. Project AB1 on VP and HP. Since AB1 is parallel to VP, $a'b_1'$, the projection of AB1 on VP is in true length inclined at ϕ to the XY line, and ab_1 , the projection of AB1 on HP is parallel to the XY line. Now the line is rotated back to its original position AB.

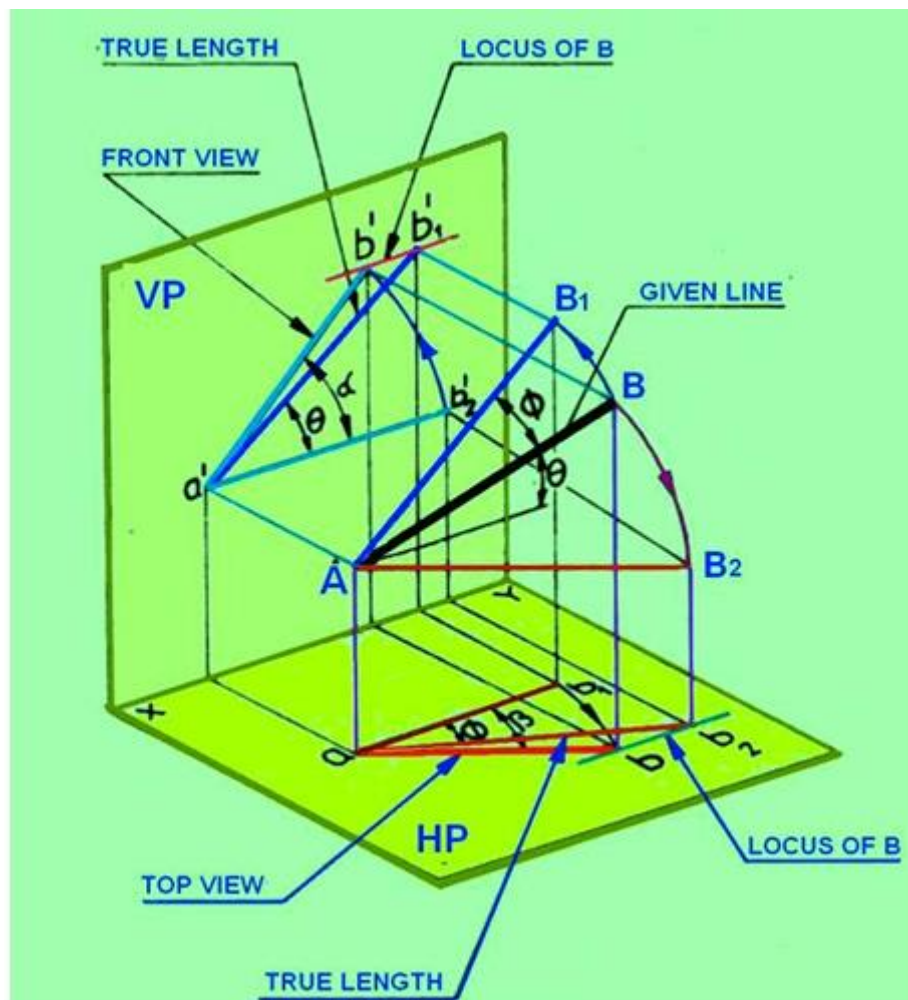


Figure 2. Illustrates the locus of end B of the line AB when the line is rotated about end A

Step 2: Rotate the line AB to make it parallel to HP

Rotate the line AB about the end A keeping ϕ the inclination of AB with VP constant, till it becomes parallel to HP as shown in figure 2. This rotation of the line will bring the end B to the second new Position B2. AB2 is the new position of the line AB, when it is inclined at ϕ to VP and parallel to HP. Project AB2 on HP and VP. Since AB2 is parallel to HP, ab_2 , the projection of AB2 on HP is in true length inclined at ϕ to XY line, and $a'b_2'$ the projection of AB2 on VP is parallel to XY line. Now the line is rotated back to its original position AB.

Step 3: Locus of end B in the front view

Referring to figure 2, when the line AB is swept around about the end A by one complete rotation, while keeping θ the inclination of the line with the HP constant, the end B will always be at the same vertical height above HP, and the locus of the end B will be a circle which appears in the front view as a horizontal line passing through b' .

As long as the line is inclined at θ to HP, whatever may be the position of the line (i.e., whatever may be the inclination of the line with VP) the length of the top view will always be equal to ab_1 and in the front view the projection of the end B lies on the locus line passing through b_1' .

Thus ab_1 , the top view of the line when it is inclined at θ to HP and parallel to VP will be equal to ab and b' , the projection of the end B in the front view will lie on the locus line passing through b_1' .

Step 4: Locus of end B in the top view

It is evident from figure 2, that when the line AB is swept around about the end A by one complete rotation, keeping the inclination of the line with the VP constant, the end B will always be at the same distance in front of VP and the locus of the end B will be a circle which appears in the top view as a line, parallel to XY, passing through b.

As long as the line is inclined at ϕ to VP, whatever may be the position of the line (i.e., whatever may be the inclination of the line with HP), the length of the front view will always be equal to $a'b_2'$ and in the top view the projection of the end B lies on the locus line passing through b_2 .

Thus $a'b_2'$ the front view of the line when it is inclined at ϕ to VP and parallel to HP, will be equal to $a'b'$ and also b, the projection of the end B in the top view lies on the locus line passing through b_2 .

Step 5: To obtain the top and front views of AB

From the above two cases of rotation it can be said that

- (i) the length of the line AB in top and front views will be equal to ab_1 and $a'b_2'$ respectively and
- (ii) The projections of the end B, (i.e., b and b') should lie along the locus line passing through b_2 and b_1' respectively.

With center a, and radius ab_2 draw an arc to intersect the locus line through b_2 at b. Connect ab the top view of the line AB

Similarly with center a' , and radius $a'b_2'$ draw an arc to intersect the locus line through b_1' at b' . Connect $a'b'$ the front view of the line AB.

Orthographic projections

As the location of one of the end points (i.e. A) with respect to HP and VP, is given, mark a and a' , the top and the front views of point A. If the line AB is assumed to be made parallel to VP and inclined at θ to HP. The front view of the line will be equal to the true length and true inclination of the line with HP. Draw $a'b_1'$ passing through a' at θ to XY line and equal to the true length of AB. $a'b_1'$ is projected down to get ab_1 , the top view parallel to the XY line. This is illustrated in figure 3.

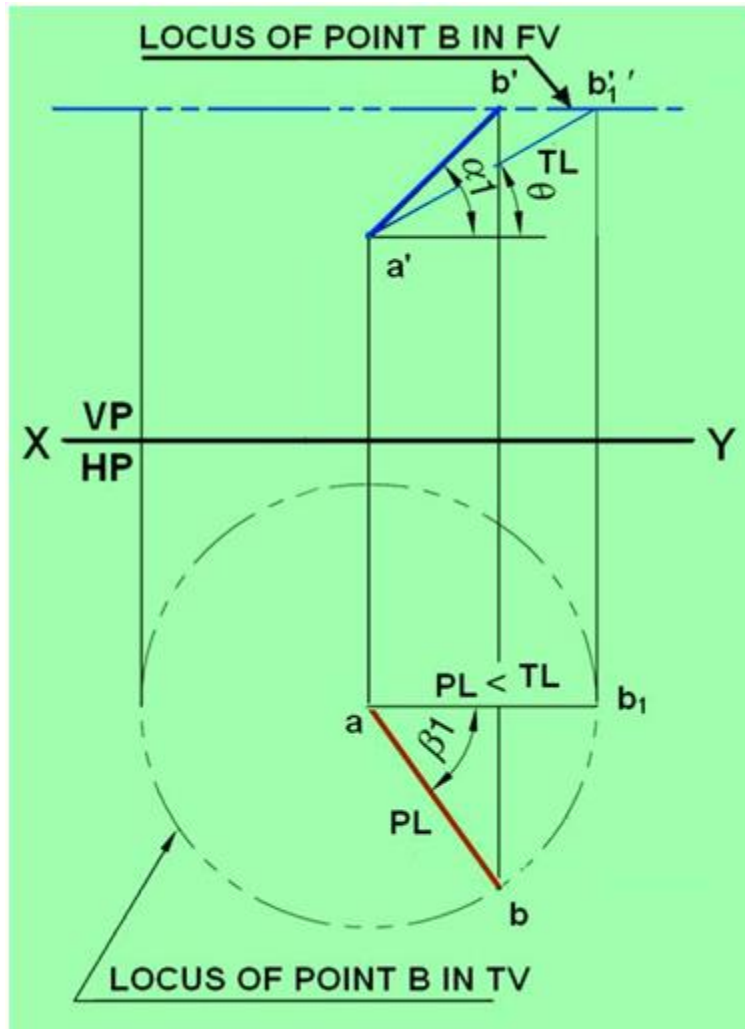


Figure 3. Illustrates the true length and true inclination of the line when it is made parallel to VP.

Now the line AB is assumed to be made parallel to HP and inclined at ϕ to VP. This is shown in figure 4. The top view of the line will be equal to the true length of the line and also ϕ , the inclination of the line with VP is seen in the top view. For this, draw ab_2 passing through a and incline at ϕ to the XY line. The length ab_2 is equal to the true length of AB. The end points a and b_2 are projected on to a line parallel to XY line and passing through a' to get $a'b_2'$ which is the front view of the line when it is parallel to HP and inclined to VP. Draw the horizontal locus lines through b_2 , and b_1' . With center a and radius ab_1 , draw an arc to cut the locus line drawn through b_2 at b . Connect ab , the top view of the line AB. With center a' and radius $a'b_2'$, draw an arc to cut the locus line drawn through b_1' at b' . Connect $a'b'$, the front view of the line AB. Orthographic projections of line AB inclined to both VP and HP, illustrating the projected length, true lengths apparent inclinations and true inclinations are shown in figure 5.

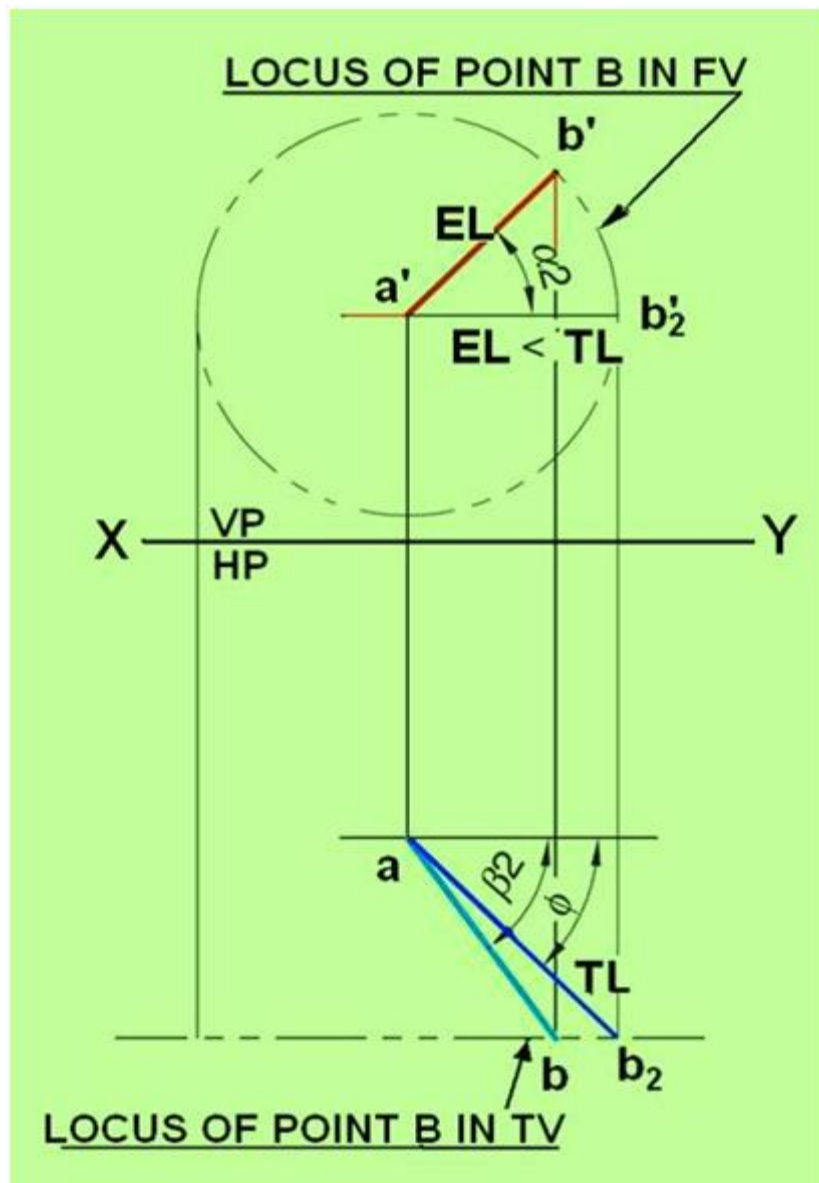


Figure 4. Illustrates the true length and true inclination of the line when it is made parallel to HP.

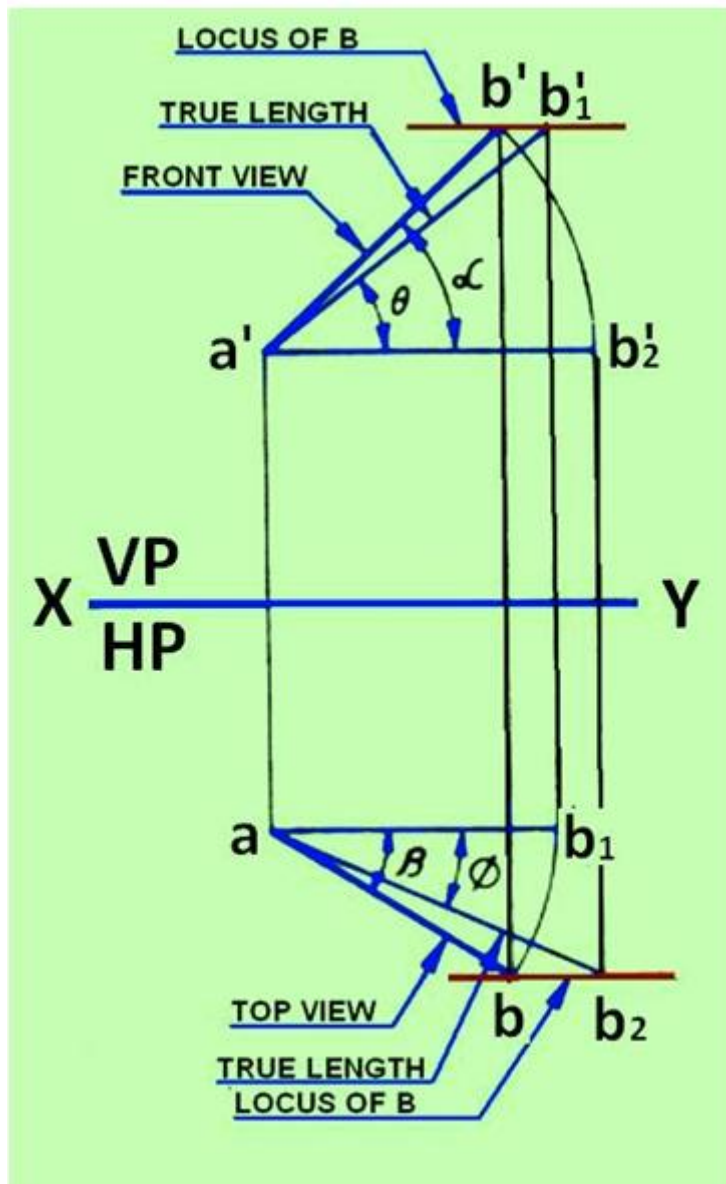


Figure 5. Illustrates the true length, apparent lengths, true inclination and apparent inclination of the line AB inclined to HP and VP.

To find true length and true inclinations of a line Many times if the top and front views of a line are given, the true length and true inclinations of a line is required to be determined.

The top and front views of the object can be drawn from if any of the following data are available:

- Distance between the end projections
- Distance of one or both the end projections.
- Apparent inclinations of the line

The problems may be solved by

- 1) Rotating line method or
- 2) Rotating trapezoidal plane method or
- 3) Auxiliary plane method

Rotating line method

The method of obtaining the top and front views of a line, when its true length and true inclinations are given. When a view of a line is parallel to the XY line, its other view will be in true length and at true inclination. By following the procedure mentioned previously, in the reverse order, the true length and true inclinations of a line from the given set of top and front views can be found. The step by step procedure is shown below in figure 1.

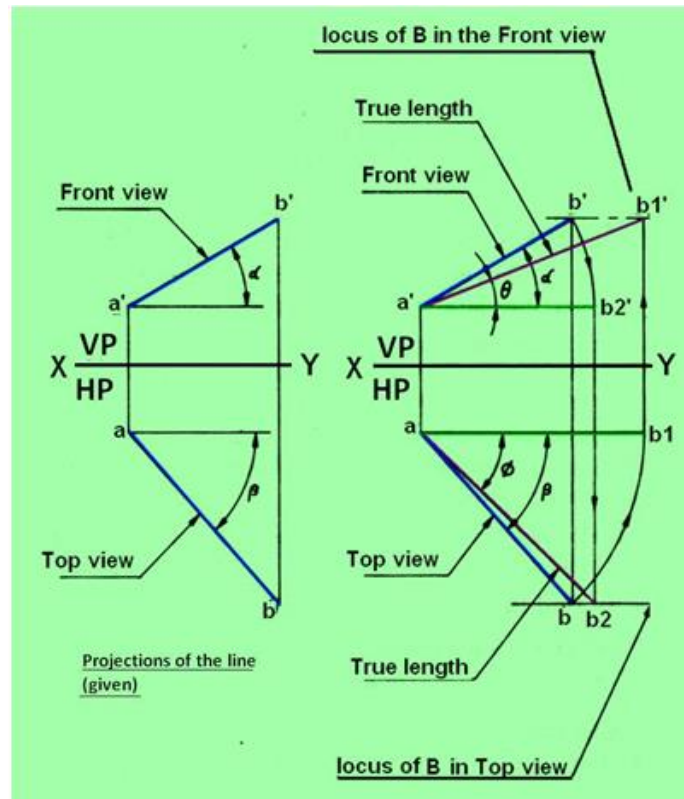


Figure 1. Determination of true length and true inclinations of a line.

- Draw the top view ab and the front view $a'b'$ as given
- Rotation of the top view: With center a and radius ab rotate the top view to the new position ab_1 to make it parallel to the XY line. Since ab_1 is parallel to the XY line, its corresponding front view will be in true length and at true inclination.
- Rotation of the front view: With center a' and radius $a'b'$ rotate the front view to the new position $a'b_1'$ parallel to the XY line. Since $a'b_1'$ is parallel to the XY line, its corresponding top view will be in true length and at true inclination. In this position, the line will be parallel to HP and inclined at θ to VP. Through b draw the locus of B in the top view. Project b_1' to get b_1 in the top view. Connect ab_1 which will be in true length and true inclination θ which the given line AB makes with VP.

Traces of a line

- The trace of a line is defined as a point at which the given line, if produced, meets or intersects a plane.
- When a line meets HP, (or if necessary on the extended portion-of HP), the point at which the line meets or intersects the horizontal plane, is called horizontal trace (HT) of the line and denoted by the letter H.
- When a line meets VP (or if necessary on the extended portion of VP), the point at which the line meets or intersects the vertical plane, is called vertical trace (VT) of the line and denoted by the letter V.

- When the line is parallel to both HP and VP, there will be no traces on the said planes. Therefore the traces of lines are determined in the following positions of the lines.
- Trace of a line perpendicular to one plane and parallel to the other.

Since the line is perpendicular to one plane and parallel to the other, the trace of the line is obtained only on the plane to which it is perpendicular, and no trace of the line is obtained on the other plane to which it is parallel. Figures 2 and 3 illustrates the trace of a line parallel to VP and perpendicular to HP and parallel to HP and perpendicular to VP respectively.

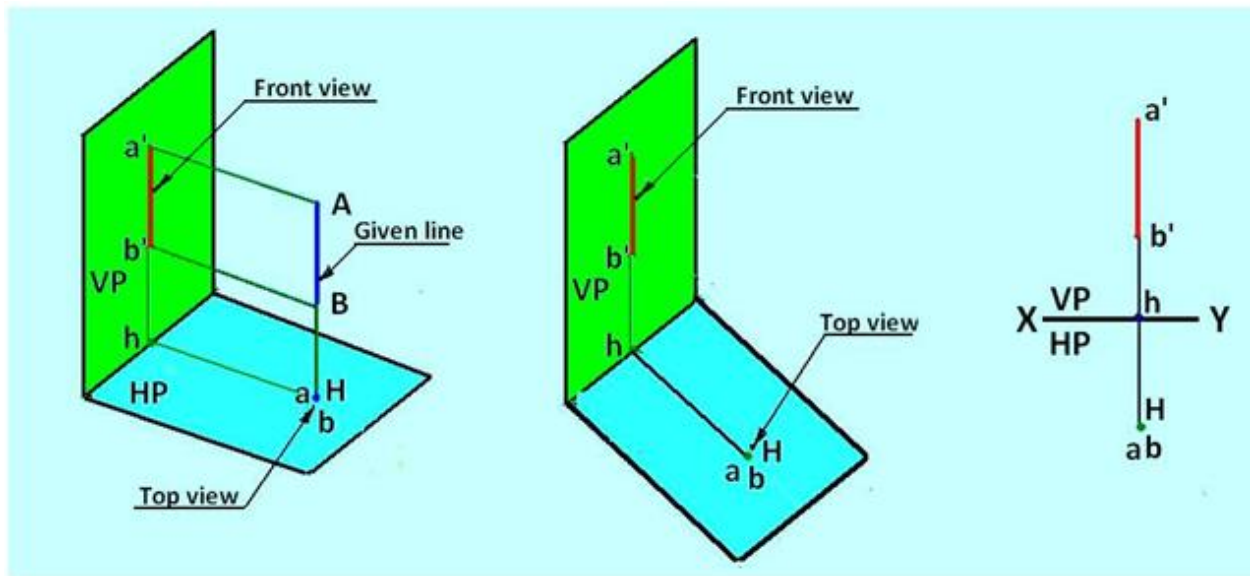


Figure 2. Trace of line parallel to VP and perpendicular to HP

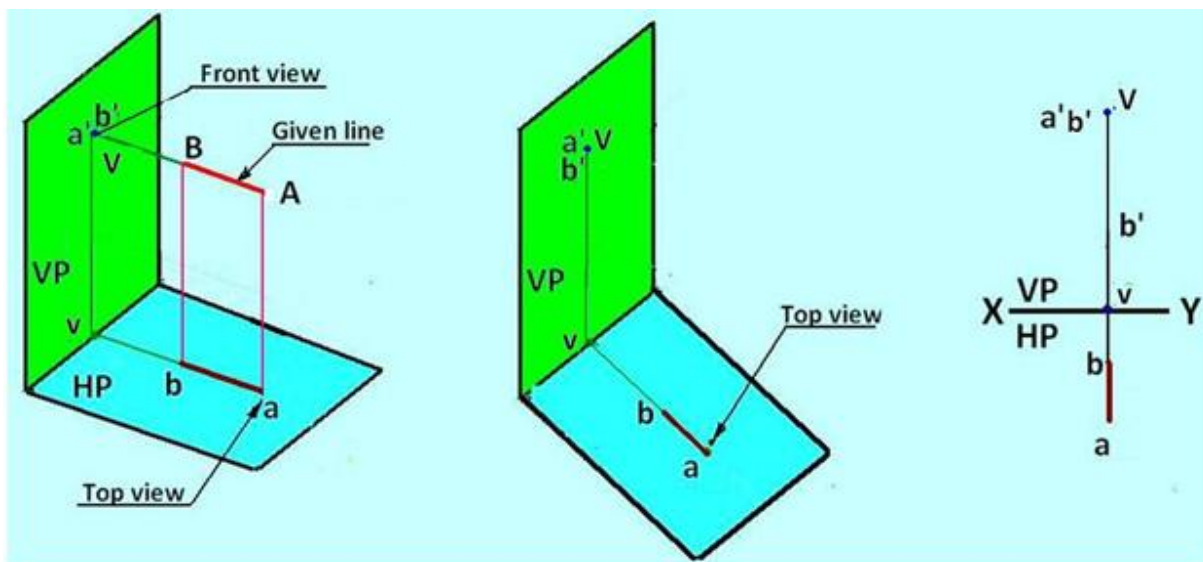


Figure 3. Trace of a line perpendicular to the VP and parallel to HP

Obtained on the plane to which it is parallel. Figure 4 shows the horizontal trace of line AB which is in lined HP and parallel to VP Traces of a line inclined to one plane and parallel to the other
When the line is inclined to one plane and parallel to the other, the trace of the line is obtained only on the plane to which it is inclined, and no trace is

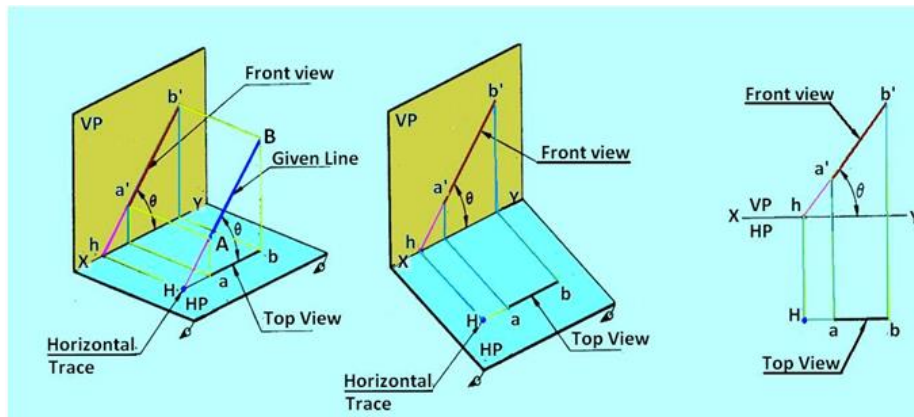


Figure 4 Horizontal trace of line AB

Figure 5 shows the vertical trace of line AB which is inclined to VP and parallel to HP

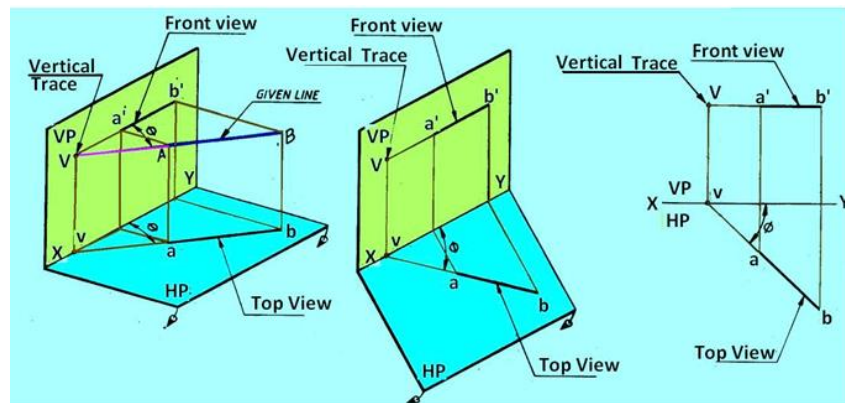


Figure 5 Vertical trace of line AB

Traces of a line inclined to both the planes

Figure 6 shows the Vertical trace (V) and Horizontal Trace (H) of Line AB inclined at α to HP and ϕ to VP

The line when extended intersects HP at H, the horizontal trace, but will never intersect the portion of VP above XY line, i.e. within the portion of the VP in the 1st quadrant. Therefore VP is extended below HP such that when the line AB is produced it will intersect in the extended portion of VP at V, the vertical trace.

In this case both horizontal trace (H) and Vertical Trace (V) of the line AB lie below XY line.

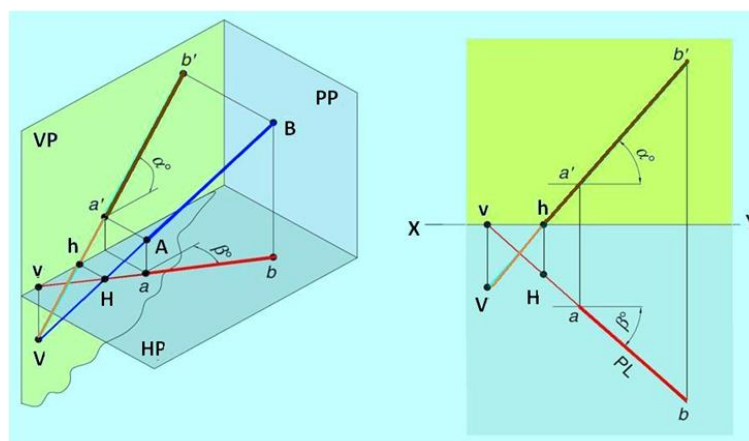


Figure 6 Vertical trace and horizontal trace of line AB which is inclined to both vertical plane and horizontal plane

Projections on Auxiliary Planes

Sometimes none of the three principal orthographic views of an object show the different edges and faces of an object in their true sizes, since these edges and faces, are not parallel to any one of the three principal planes of projection. In order to show such edges and faces in their true sizes, it becomes necessary to set up additional planes of projection other than the three principal planes of projection in the positions which will show them in true sizes. If an edge or a face is to be shown in true size, it should be parallel to the plane of projection. Hence the additional planes are set up so as to be parallel to the edges and faces which should be shown in true sizes. These additional planes of projection which are set up to obtain the true sizes are called Auxiliary Planes. The views projected on these auxiliary planes are called Auxiliary Views.

The auxiliary view method may be applied

- To find the true length of a line.
- To project a line which is inclined to both HP and VP as a point.
- To project a plane surface or a lamina as a line.

Types of auxiliary planes

Usually the auxiliary planes are set up such that they are parallel to the edge or face which is to be shown in true size and perpendicular to any one of the three principal planes of projection. Therefore, the selection of the auxiliary plane as to which of the principal planes of projection it should be perpendicular, obviously depends on the shape of the object whose edge or face that is to be shown in true size.

- If the auxiliary plane selected is perpendicular to HP and inclined to VP, the view of the object projected on the auxiliary plane is called auxiliary front view and the auxiliary plane is called auxiliary vertical plane and denoted as AVP.
- If the auxiliary plane is perpendicular to VP and inclined to HP, the view of the object projected on the auxiliary plane is called auxiliary top view and the auxiliary plane is called auxiliary inclined plane and denoted as AIP.

Auxiliary Vertical (AVP) plane

- An AVP is placed in the first quadrant with its surface perpendicular to HP and inclined at Φ to VP. The object is assumed to be placed in the space in between HP, VP and AVP. The AVP intersects HP along the X_1Y_1 line. The direction of sight to project the auxiliary front view will be normal to AVP. The position of the auxiliary vertical plane w.r.t HP and VP is shown in figure 1.
- After obtaining the top view, front view and auxiliary front view on HP, VP and AVP, the HP, with the AVP being held perpendicular to it, is rotated so as to be in-plane with that of VP, and then the AVP is rotated about the X_1Y_1 line so as to be in plane with that of already rotated HP.

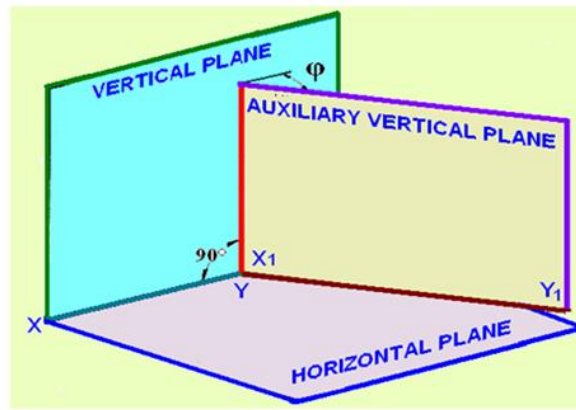


Figure 1. The position of the auxiliary vertical plane w.r.t HP and VP

Auxiliary Inclined Plane (AIP)

- AIP is placed in the first quadrant with its surface perpendicular to VP and inclined at θ to HP. The object is to be placed in the space between HP, VP and AIP. The AIP intersects the VP along the X_1Y_1 line. The direction of sight to project the auxiliary top view will be normal to the AIP. The position of the AIP w.r.t HP and VP is shown in figure 2.
- After obtaining the top view, front view and auxiliary top view on HP, VP and AIP, HP is rotated about the XY line independently (detaching the AIP from HP). The AIP is then rotated about X_1Y_1 line independently so as to be in-plane with that of VP.

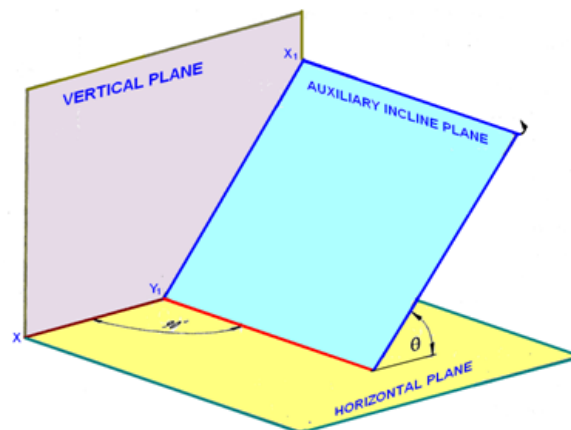


Figure 2. The position of the AIP w.r.t HP and VP

Projection of Points on Auxiliary Planes

- Projection on Auxiliary Vertical Plane
- Point P is situated in the first quadrant at a height m above HP. An auxiliary vertical plane AVP is set up perpendicular to HP and inclined at Φ to VP. The point P is projected on VP, HP and AVP. As shown in figure 3, p' is the projection on VP, p is the projection on HP and P_1' is the projection on AVP.
Since point is at a height m above HP, both p' and P_1' are at a height m above the XY and X_1Y_1 lines, respectively

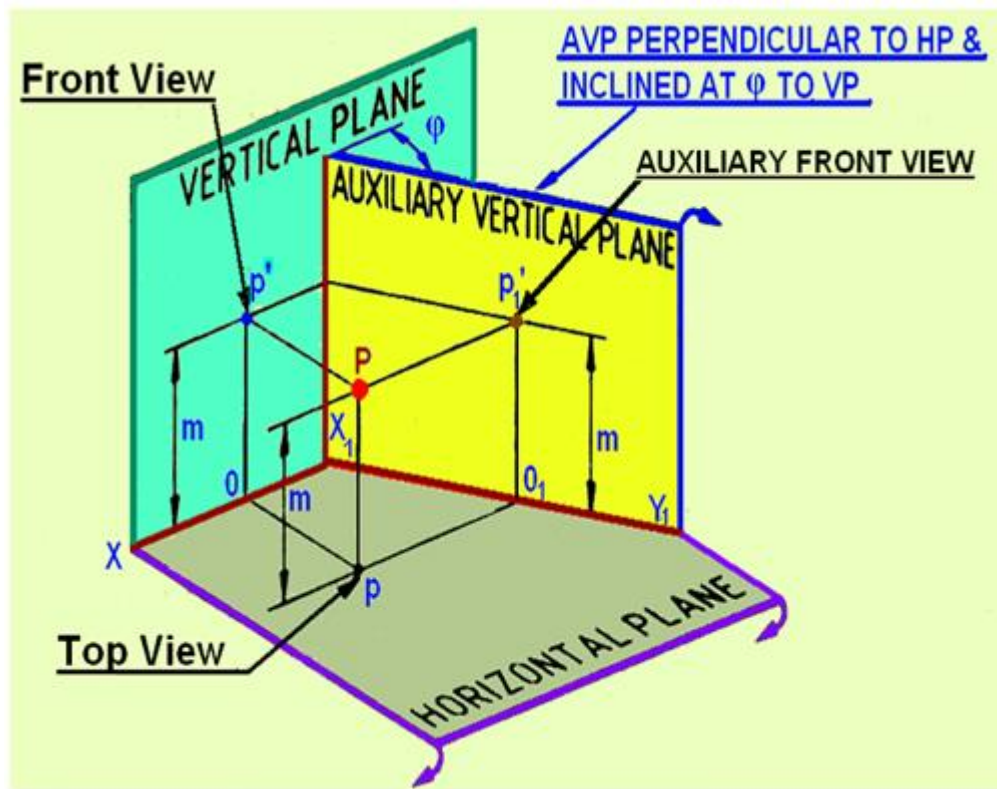


Figure 3. Projection of Point P on VP, HP and AV

HP is rotated by 90 degree to bring it in plane of VP (figure 4(a)). Subsequently, the AVP is rotated about the X_1Y_1 line (figure 4(b)), such that it becomes in-plane with that of both HP and VP.

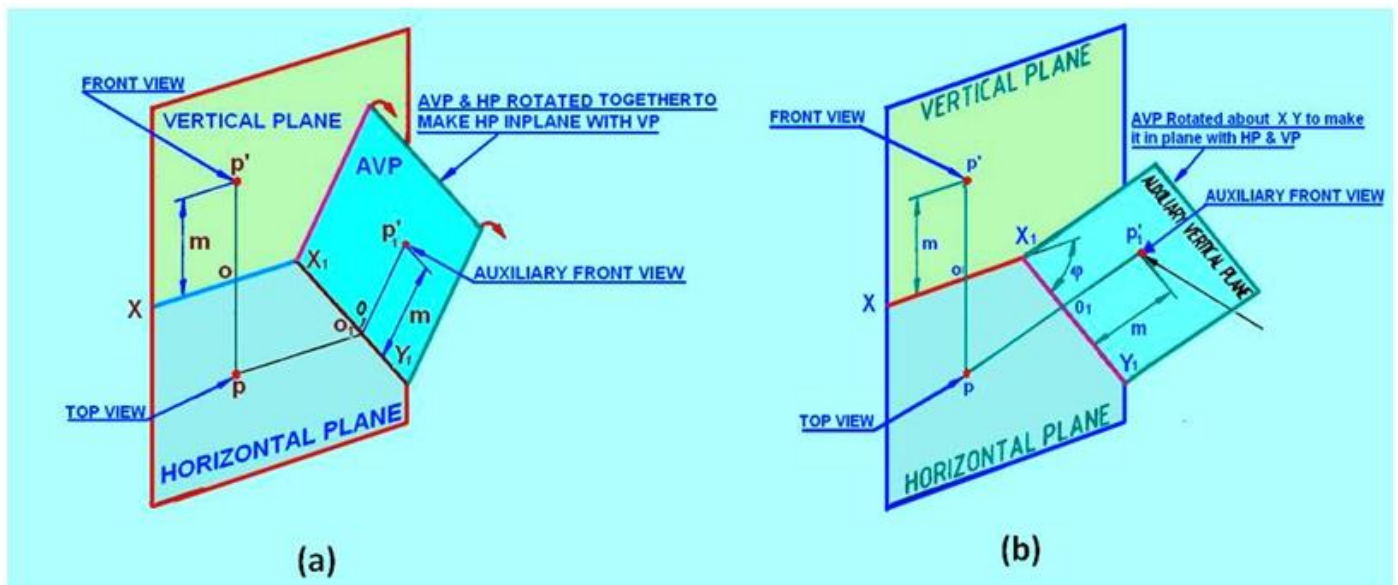


Figure 4. The rotation of (a) HP and (b) AVP to make HP and AVP in plane with VP.

The orthographic projections (projections of point P on HP, VP and AVP) of point P can be obtained by the following steps.

Draw the XY line and mark p and p', the top and front views of the point P. Since AVP is inclined at Φ to VP, draw the X_1Y_1 line inclined at Φ to the XY line at any convenient distance from p. Since point P is at a height m above HP, the auxiliary front view p1' will also be at a height m above the

X1Y1 line. Therefore, mark P1' by measuring $o_1p_1' = op' = m$ on the projector drawn from p perpendicular to the X1Y1 line.

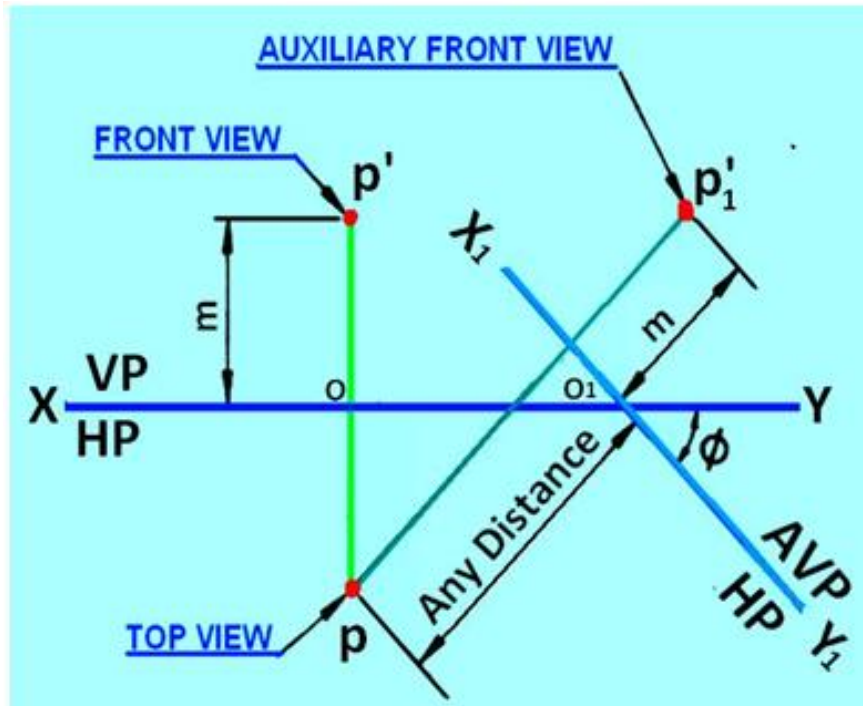


Figure 5. Orthographic projection of the point P by Auxiliary projection method.

Projection on Auxiliary Inclined Plane

Point P is situated in first quadrant at a distance n from VP. An auxiliary plane AIP is set up perpendicular to VP and inclined at θ to HP. The point P is projected on VP, HP and AIP.

p' is the projection on VP, p is the projection on HP and P_1 is the projection on AIP. Since the point is at a distance n from VP, both p and p_1 are at a distance n above the XY and X1Y1 lines, respectively

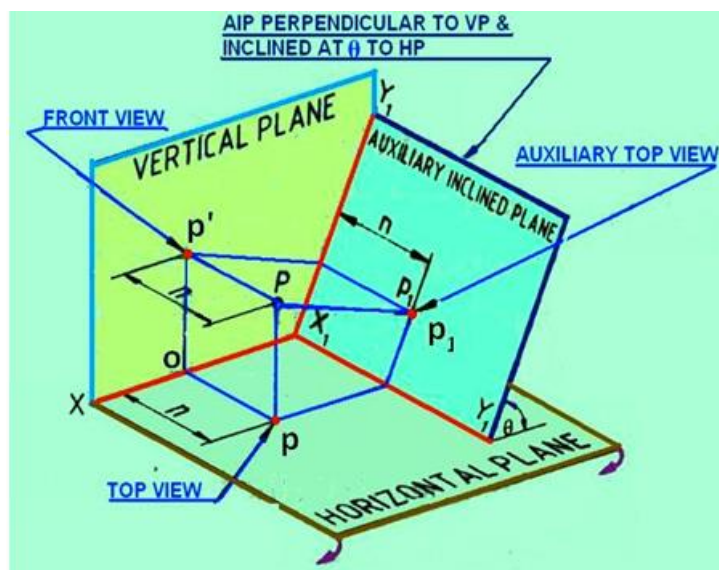


Figure 6. Orthographic projection of point P by auxiliary projection on AIP.

HP is now rotated by 90° about XY line to bring it in plane with VP, as shown in figure 7(a). After the HP lies in-plane with VP, the AIP is rotated about the X_1Y_1 line, so that it becomes in-plane with that of both HP and VP.

p and p' lie on a vertical projector perpendicular to the XY line, and p' and p_1 lie on a projector perpendicular to the X_1Y_1 line which itself is inclined at θ to XY line.

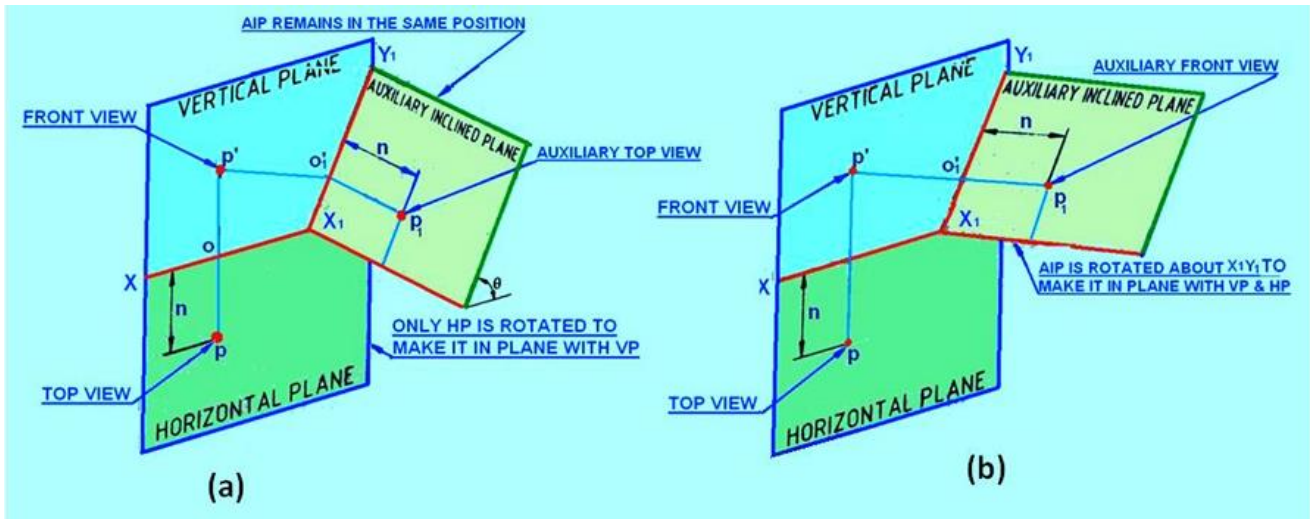
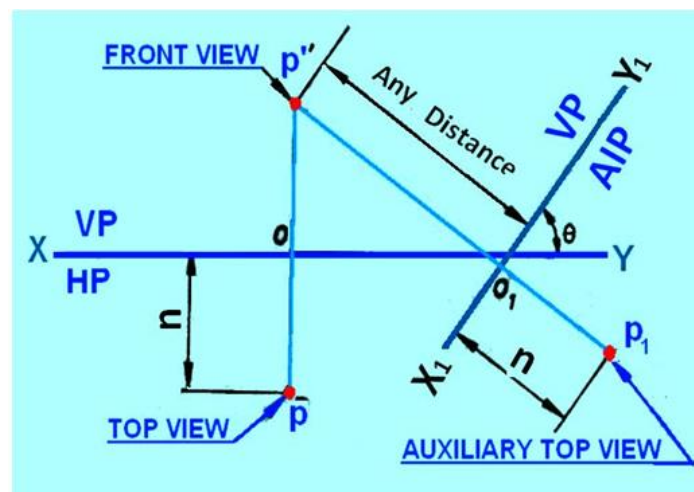


Figure 7. Orthographic projection of point P by auxiliary projection on AIP.

The orthographic projections (projections of point P on HP, VP and AIP) of point P can be obtained by the following steps.

- Draw the XY line and mark p and p' , the top and front views of the point P.
- Since AIP is inclined at q to HP, draw the X_1Y_1 line inclined at q to the XY line at any convenient distance from p' .
- Since point P is at a distance n in front of VP, the auxiliary top view p_1 will also be at a distance n from the X_1Y_1 line.
- Therefore, mark P_1 by measuring $o_1p_1 = op = n$ on the projector drawn from p' perpendicular to the X_1Y_1 line.



Step by step procedure to draw auxiliary views

Auxiliary front view	Auxiliary top view
Draw the top and front views.	Draw the top and front views
Draw X_1Y_1 line inclined at ϕ (the inclination of AVP with VP) to the XY line.	Draw X_1Y_1 line inclined at ψ (the inclination of AIP with HP) to XY line.
Draw the projectors through the top views of the points perpendicular to the X_1Y_1 line.	Draw the projectors through the front views of the points perpendicular to the X_1Y_1 line.
The auxiliary front view of a point is obtained by stepping off a distance from the X_1Y_1 line equal to the distance of the front view of the given point from the XY line.	The auxiliary top view of a point is obtained by stepping off a distance from X_1Y_1 line equal to the distance of the top view of the given point from the XY line

A: Plane surface parallel to one plane and perpendicular to the other two

Consider A triangular lamina placed in the first quadrant with its surface parallel to VP and perpendicular to both HP and left PP. The lamina and its projections on the three projection planes are shown in figure

$a'b'c'$ is the front view, abc the top view and $a''b''c''$ the side view
 Since the plane is parallel to VP, the front view $a'b'c'$ shows the true shape of the lamina. Since the lamina is perpendicular to both HP and PP, the top view and side views are seen as lines.

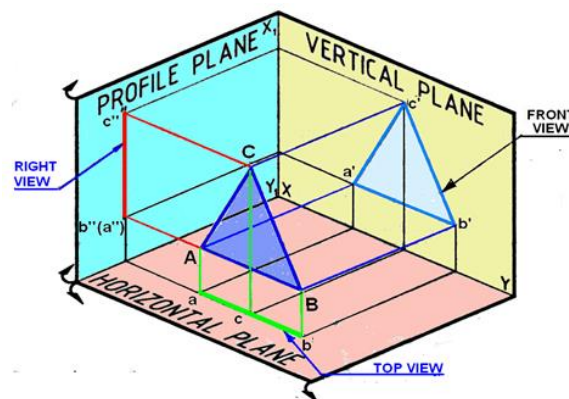


Figure 1. Projections of a triangular lamina on the projection planes

After projecting the triangular lamina on VP, HP and PP, both HP and PP are rotated about XY and X_1Y_1 lines, as shown in figure 2, till they lie in-plane with that of VP

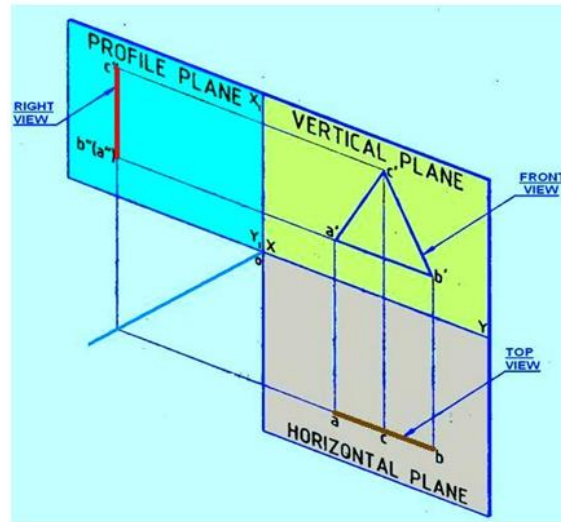


Figure 2. Rotation of PP and HP after projection.

The orthographic projections of the plane, shown in figure 3 can be obtained by the following steps.

Draw XY and X₁Y₁ lines and mark HP, VP and left PP. Draw the triangle $a'b'c'$ in true shape to represent the front view at any convenient distance above the XY line. In the top view the triangular lamina appears as a line parallel to the XY line. Obtain the top view acb as a line by projecting from the front view at any convenient distance below the XY line.

Since the triangular lamina is also perpendicular to left PP, the right view will be a *line* parallel to the X₁Y₁ line. To project the right view, draw a 45° line at the point of intersection of the XY and X₁Y₁ lines. Draw the horizontal projector through the corner a in the top view to cut the 45° line at m . Through m draw a vertical projector. From the corners c' and a' in the front view draw the horizontal projectors to cut the vertical projector drawn through m at c'' and b'' . In the right view the corner A coincides with B and hence is invisible.

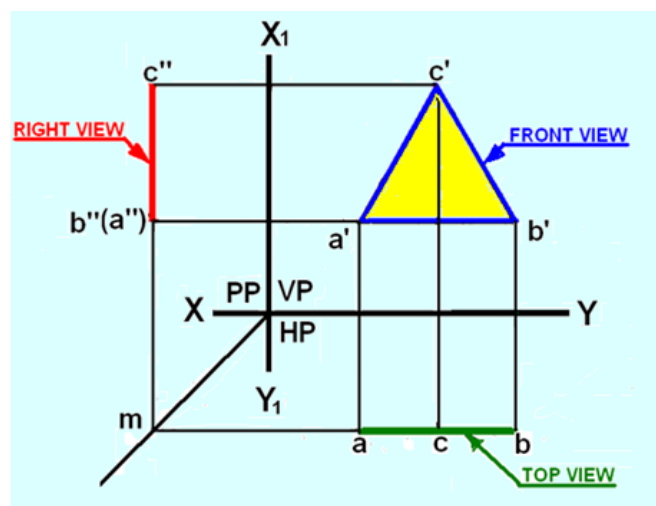


Figure 3. Orthographic projections of the lamina ABC

B) Plane parallel to HP and perpendicular to both VP and PP

A square lamina (plane surface) is placed in the first quadrant with its surface parallel to HP and perpendicular to both VP and left PP. Figure 4 (a) shows the views of the object when projected on to the three planes. Top view is shown as $abcd$, the front view as $a' (d') b' (c')$ and the side view as $b'' (a'') c'' (d'')$. Since

the plane is parallel to the HP, its top view $abcd$ will be in its true shape. Since the plane is perpendicular to VP and PP, its front and side views will be lines $a' (d') b' (c')$ and $b'' (a'') c'' (d'')$ respectively. After projecting the square lamina on VP, HP and PP, both HP and PP are rotated about XY and X_1Y_1 lines, as shown in figure 4(b), till they lie in-plane with that of VP.

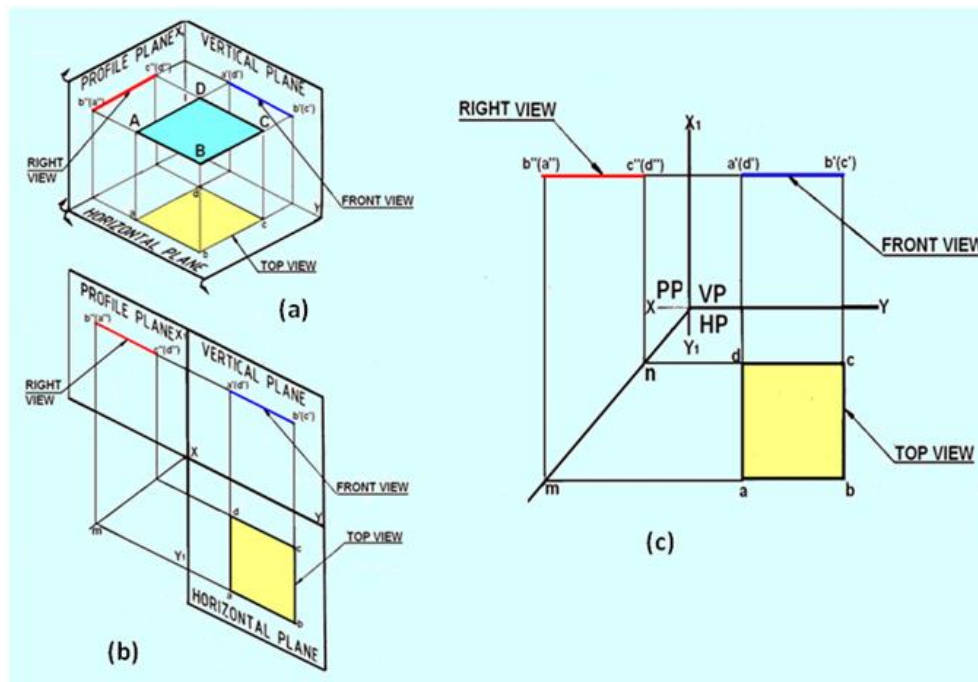


Figure 4. Projections of the lamina with its surface parallel to HO and perpendicular to both VP and PP.

The orthographic projections of the plane, shown in figure 4(c) can be obtained by the following steps.

- Draw XY and X_1Y_1 lines and mark HP, VP and left PP.
- Draw the square $abcd$ in true shape to represent the top view at any convenient distance below the XY line.
- In the front view, the square lamina appears as a line parallel to the XY line.
- Obtain the front view as a line $a'(d')b'(c')$ by projecting from the top view, parallel to the XY line at any convenient distance above it.
- In the front view, the rear corners D and C coincide with the front corners A and B, hence d' and c' are indicated within brackets.
- Since the square lamina is also perpendicular to left PP, the right view projected on it will also be a line perpendicular to X_1Y_1 line.
- Project the right view as explained in the previous case.
- In right view, the corners A and D coincide with the corners B and C respectively, hence (a') and (d') , are indicated within brackets.

C) Plane parallel to PP and perpendicular to both HP and VP

A pentagon lamina (plane surface) is placed in the first quadrant with its surface is parallel to left PP and perpendicular to both VP and HP.

Figure 5 (a) shows the views of the object when projected on to the three planes.

- Side view is shown as $a''b''c''d''e''$, the front view as $b'(c')a'(d')e'$ and the top view as $a(b)e(c)d$.
- Since the plane is parallel to the PP, its side view $a''b''c''d''e''$ will be in its true shape.
- Since the plane is perpendicular to VP and HP, its front and side views will be projected as lines.
- After projecting the pentagonal lamina on VP, HP and PP, both HP and PP are rotated about XY and X₁Y₁ lines, as shown in figure 5(b), till they lie in-plane with that of VP.

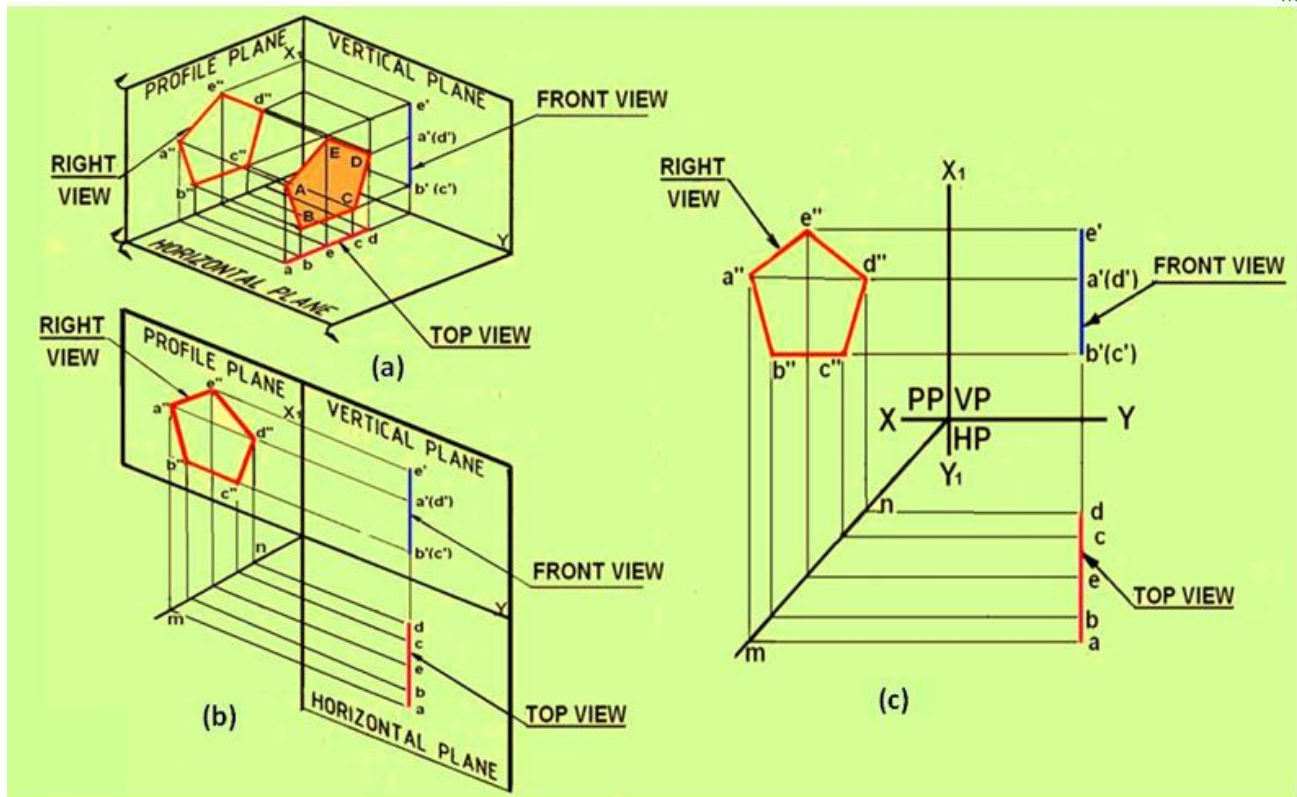


Figure 5 Projections of a pentagonal lamina with its surface parallel to PP and perpendicular to HP and VP.

The orthographic projections of the plane, shown in figure 5(c) can be obtained by the following steps.

- Draw XY and X₁Y₁ lines, and mark HP, VP and left PP.
- Draw the pentagon $a''b''c''d''e''$ in true shape to represent the side view at any convenient distance above the XY line and left of X₁Y₁ line.
- The top and front views of the lamina appear as lines perpendicular to XY line.
- Obtain the front view $b'(c')a'(d')e'$ as a line by projecting from the right view at any convenient distance from the X₁Y₁ line.
- In the front view, the rear corners D and C coincide with A and B respectively, hence d' and c' are indicated within brackets.
- The orthographic projections of the plane, shown in figure 4(c) can be obtained by the following step.
- Since the pentagon lamina is also perpendicular to HP, the top view also appears as a line.
- Project the top view from the right and front views.

D) Plane surface perpendicular to one plane and inclined to the other two Plane inclined at Φ to VP and perpendicular to HP

Draw the projections of a triangular lamina (plane surface) placed in the first quadrant with its surface inclined at ϕ to VP and perpendicular to the HP.

Since the lamina is inclined to VP, it is also inclined to left PP at $(90 - \phi)$. The triangular lamina ABC is projected onto VP, HP and left PP. $a'b'c'$ – is the front view projected on VP. $a''b''c''$ – is the right view projected on left PP.

Since lamina is inclined to VP and PP, front and side views are not in true shape. Since lamina is perpendicular to HP, its top view is projected as a line acb . Figure 6 (c) shows the multi view drawing of the lamina.

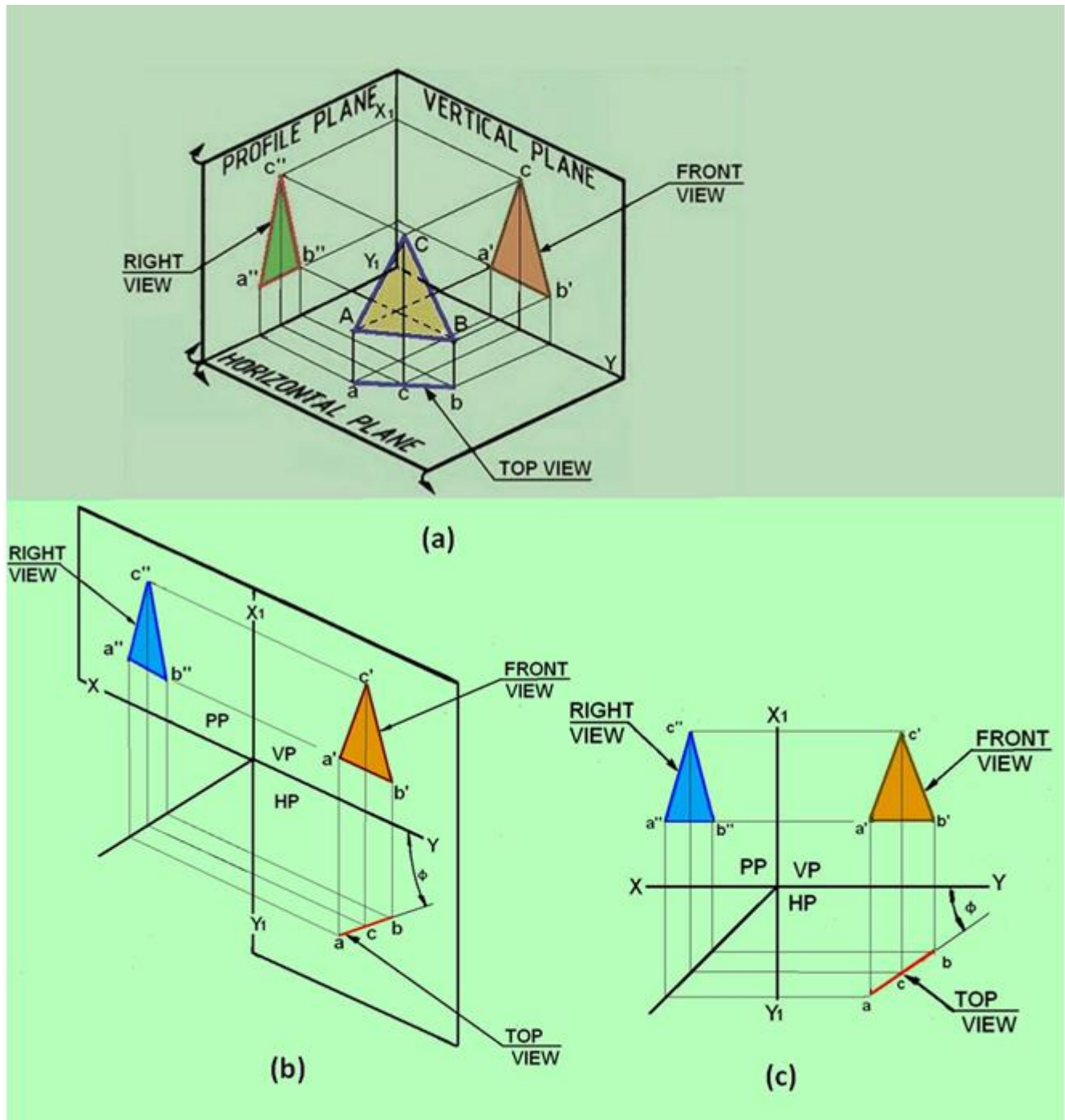


Figure 6. The projections of the triangular lamina

Problem 1: A regular pentagon lamina of 30 mm side rests on HP with its plane surface vertical and inclined at 30° to VP. Draw its top and front views when one of its sides is perpendicular to HP.

Solution: The projections of the pentagonal lamina has its surface vertical (i.e., perpendicular to HP) and inclined at 30° to VP. Since the lamina is inclined to VP, initially it is assumed to be parallel to VP. In this position one of the sides of the pentagon should be perpendicular to HP. Therefore, draw a regular pentagon $a'b'c'd'e'$ in the VP to represent the front view with its side $a'e'$ perpendicular to HP. Since the lamina is perpendicular to HP, the top view will be a line, $a(e)b(d)c$. Assume that edge $a'e'$ perpendicular to HP in the final position. The top view of the lamina is now rotated about $a(e)$ such that the line is inclined at 30° to XY line, as shown by points a_1, b_1, c_1, d_1 , and e_1 in the right bottom of Figure 1. Draw vertical projectors from points a_1, b_1, c_1, d_1 , and e_1 . Draw horizontal projectors from points a', b', c', d' , and e' . The intersection gives the respective positions of the points in the Front view. Join a_1', b_1', c_1', d_1' , and e_1' to obtain the Front view of the lamina.

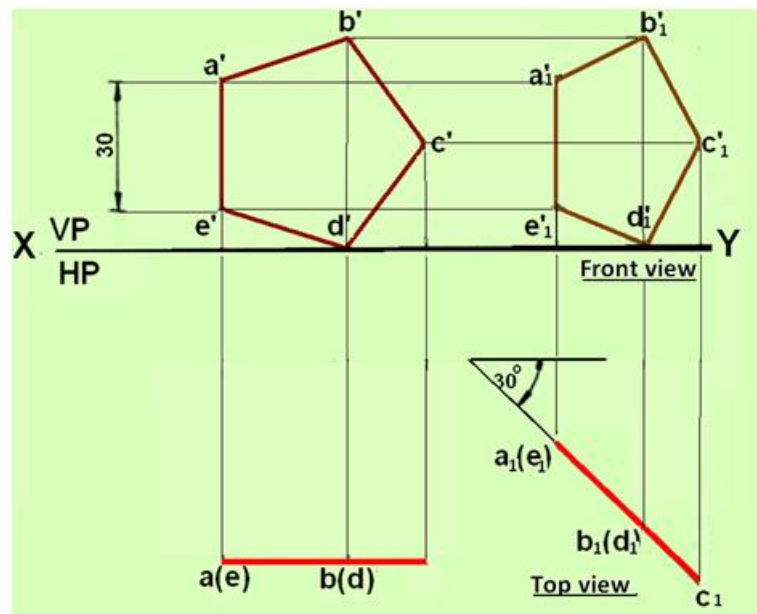


Figure 1. Orthographic projections of the pentagonal lamina.

Problem 2. Draw the front view, top view and side view of a square lamina. The surface of the lamina is inclined at θ to HP and perpendicular to VP.

Solution. The three views of the square lamina is shown in figure 2. Since the lamina is perpendicular to VP, its front view will be a line $[a'(b') c' (d')]$ having length as the true length of the edge of the square and inclined at θ to XY line. The corners B and C coincide with A and D in the front view. Since the lamina is inclined to HP at θ , it is also inclined to the left PP at $(90 - \theta)$. The square lamina is projected on to VP, HP and left PP. Draw vertical projectors from points a', b', c' and d' . On any position on these lines construct the rectangle $a-b-c-d$ such that length ab and cd are equal to the true length of the square edge. The rectangle $a-b-c-d$ is the top view of the lamina. The side view of the lamina a'', b'', c'' and d'' can be obtained by drawing projectors from points a', b', c' and d' and a, b, c , and d .

Worked out problem in Auxiliary projections

Problem 1.

The end projectors of a line AB is 40 mm. The point A is 24 mm above HP and 10 mm in front of VP. Point B is 46 mm above HP and 46 mm in front of VP. Determine the True length of the line and its True inclinations with both the reference planes.

Solution: The solution to the problem is shown in figure 1.

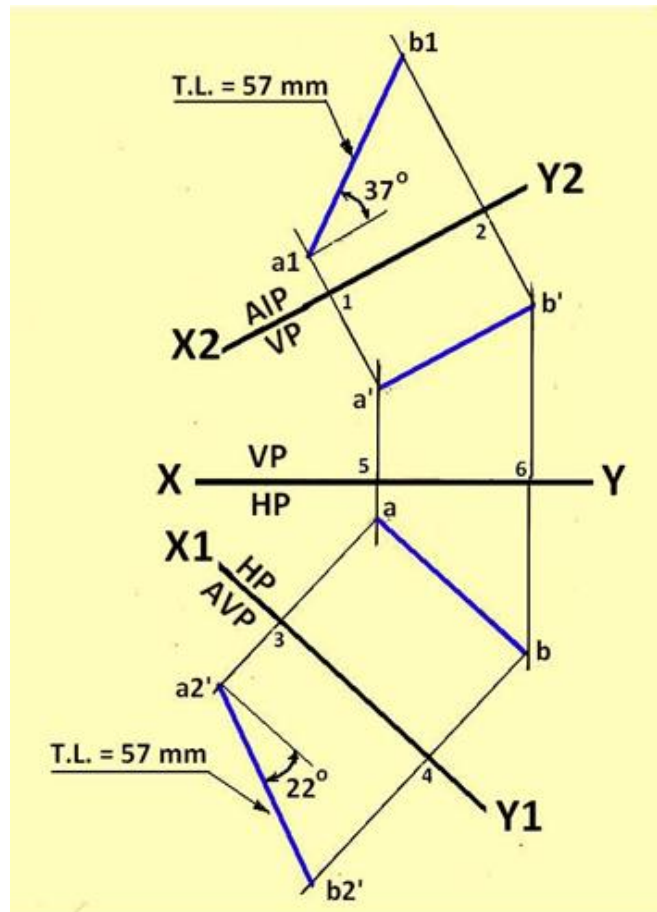


Figure1. Solution for problem No. 2.

- Draw the XY line. Draw two line perpendicular to XY line separated by a horizontal distance of 40 mm.
- Along the first line mark points a' and a 24 mm above and 10 mm below XY line.
- Along the second line mark b and b' 46 mm above and below XY line.
- Draw a reference line X1 Y1 parallel to line ab to represent the AVP.
- Through a and b, draw projectors to X1 Y1 line and extend it.
- Mark a2' such that 3-a2' = 5-a'. similarly mark b2' such that 4-b2' = 6b'.
- Join a2'-b2' to obtain the true length of the line.
- The inclination of the line a2'-b2' with X1Y1 line is the true inclination of the line with HP.
- Draw X2Y2 parallel to the front view a'b' and project the auxiliary top view a1b1 in the similar manner. The inclination of the line a1-b1 with X2Y2 line is the true inclination of the line with VP.

By measurement, the following dimensions are obtained:

- True length of the line is 57 mm
- True inclination of the line with HP = 22°
- True inclination of the line with VP = 37°

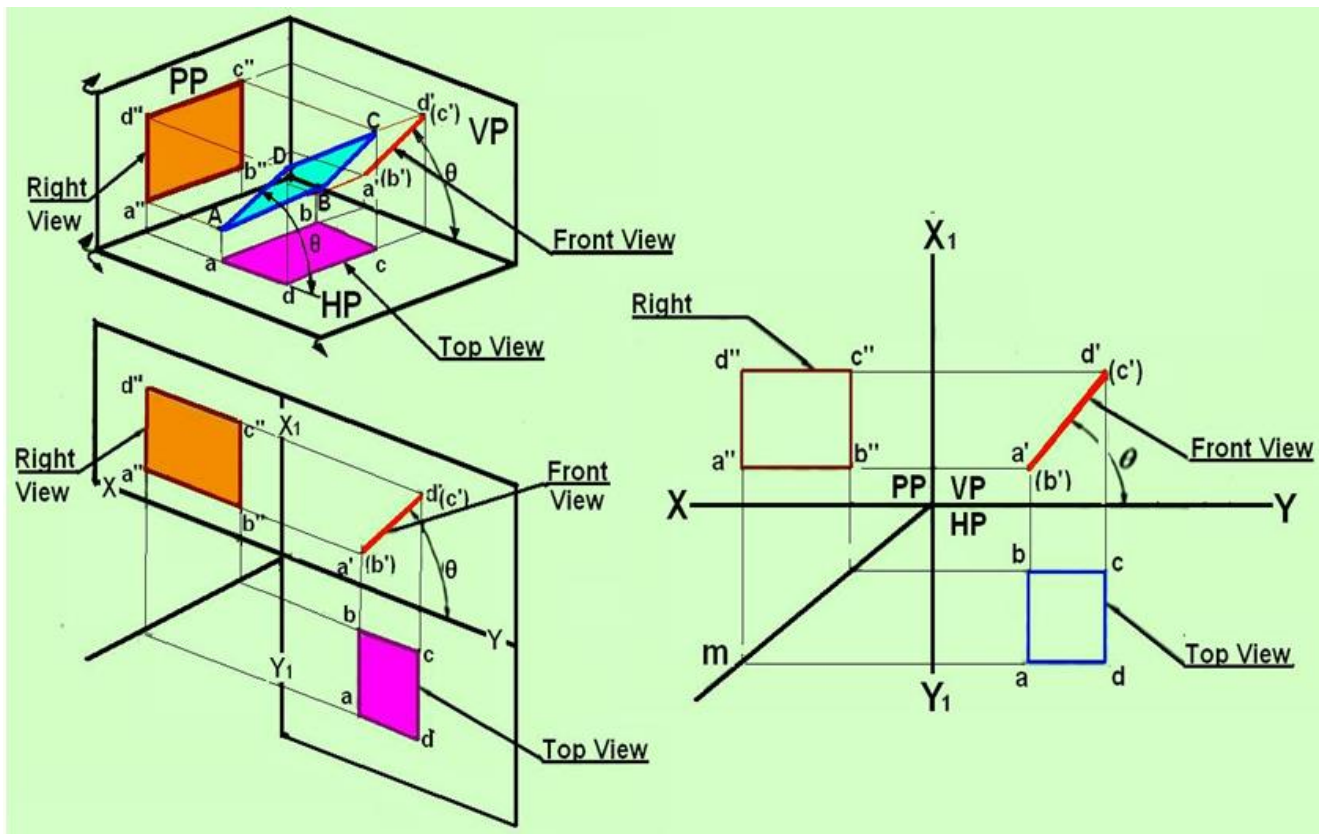


Figure 2. The projection of the square lamina as mentioned in problem 2.

Problem 3. Draw the Top view and front view of a circular lamina if the surface of the lamina is perpendicular to HP and inclined at 30° to VP.

Solution: The projections of the circular lamina are shown in figure 3.

- Let us first assume that the plane is perpendicular to HP and parallel to VP.
- The Front view will be a circle and with diameter equal to the diameter of the lamina.
- Divide the circle into 12 equal parts and label them as $1', 2', 3', \dots, 12'$.
- The top view will be a straight line $1-7$, parallel to XY line and can be obtained by drawing projectors from $1', 2', \dots$ and $12'$.
- Since the circle is inclined at 30° to VP and perpendicular to HP, reconstruct the top view such that the straight line is inclined at 30° to XY line. Let the respective points be $11, 21, 31, \dots, 121$.
- Draw vertical projectors from points $11, 21, 31, \dots, 121$ to meet the horizontal projectors from points $1', 2', 3', \dots, 12'$ to obtain the points $11', 21', 31', \dots, 121'$ in the Front view.
- Draw a smooth curve passing through points $11', 21', 31', \dots, 121'$ to obtain the Front view of the circular lamina.

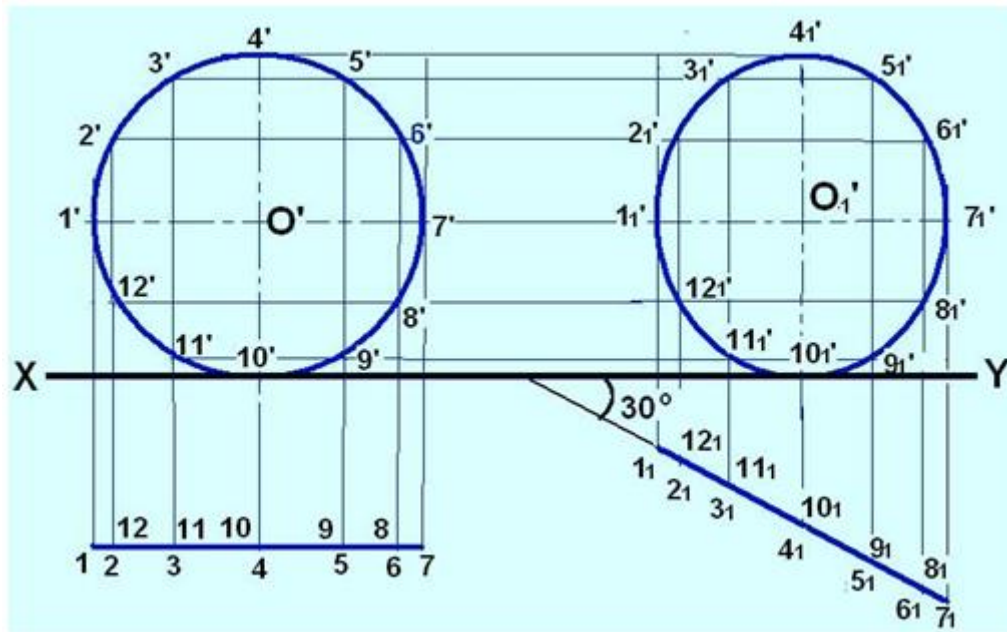


Figure 3. Projections of the circular lamina mentioned in problem 3.

To find the true shape of a plane surface

The true shape of plane surface when its top and front views are given may be determined by setting up two auxiliary planes and projecting on to these. The example below will demonstrate the method of finding the true shape of a quadrilateral.

Problem 4: The corners of a quadrilateral PQRS are as follows: P is 25 mm above HP and 50 mm in front of VP, Q is in HP and 80 mm in front of VP. R is 50 mm above HP and 40 mm in front of VP. S is 65 mm above HP and 20 mm in front of VP. The distances between the vertical projectors parallel to the XY lines are as follows: Between P and S is 20 mm, between P and Q is 35 mm, between P and R is 60 mm. Draw the top and front views of the quadrilateral and find its true shape.

Solution:

The method of obtaining the true shape of the lamina is shown in figure 4.

- First draw the front view and top view of the lamina as per the conditions mentioned in the problem.
- Through any one corner in any of the two views, say p in the top view, draw a line parallel to the XY line to intersect the edge qr at t. Project t to the top view to get t' on q'r'.
- Connect p't'.
- Since pt is horizontal, p't' is in true length.
- The point view of line p't' can be obtained by projecting on to an AIP (by drawing the reference line X1Y1 perpendicular to p't').
- Project the four corner points to get the Auxiliary top view s1 r1 p1 q1 (line view).
- Project the auxiliary Front View on to another Auxiliary vertical plane by drawing the X2Y2 line, parallel to s1r1p1q1 line.
- The Auxiliary Front view will be the true shape of the object.

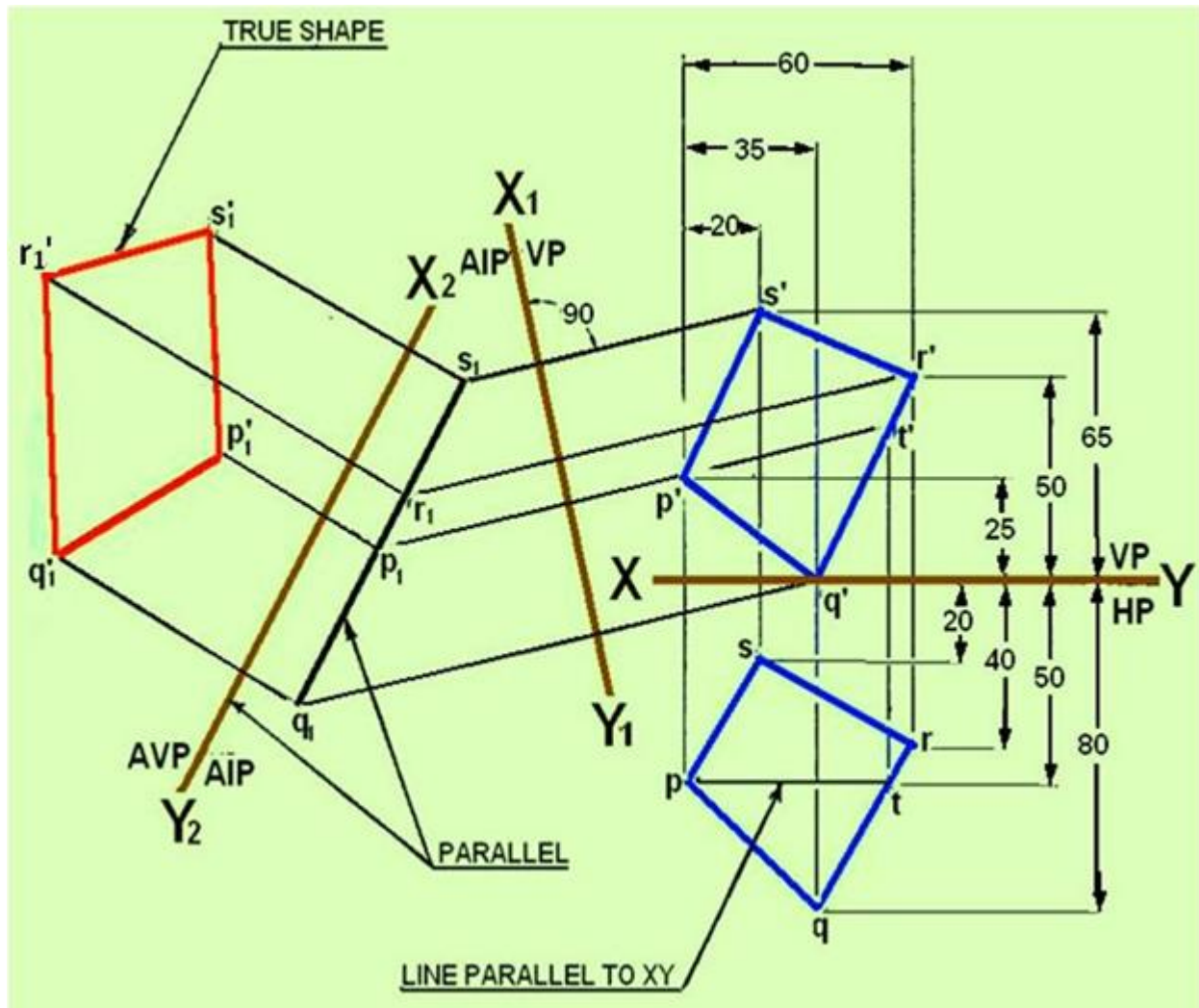


Figure 4. Solution of problem 4.

Projections of lines (Drawing practice)

Problem -1

A straight line AB of true length 100 mm has its end A 20 mm above HP and 30 mm in front of VP. The top view of the line is 80 mm and front view is 70 mm. Draw the projections (TV and FV) of the line AB and obtain the true inclinations of the line AB with HP and VP.

Solution: The solution to the problem is shown in figure 1. The step wise procedure for the solution is discussed below:

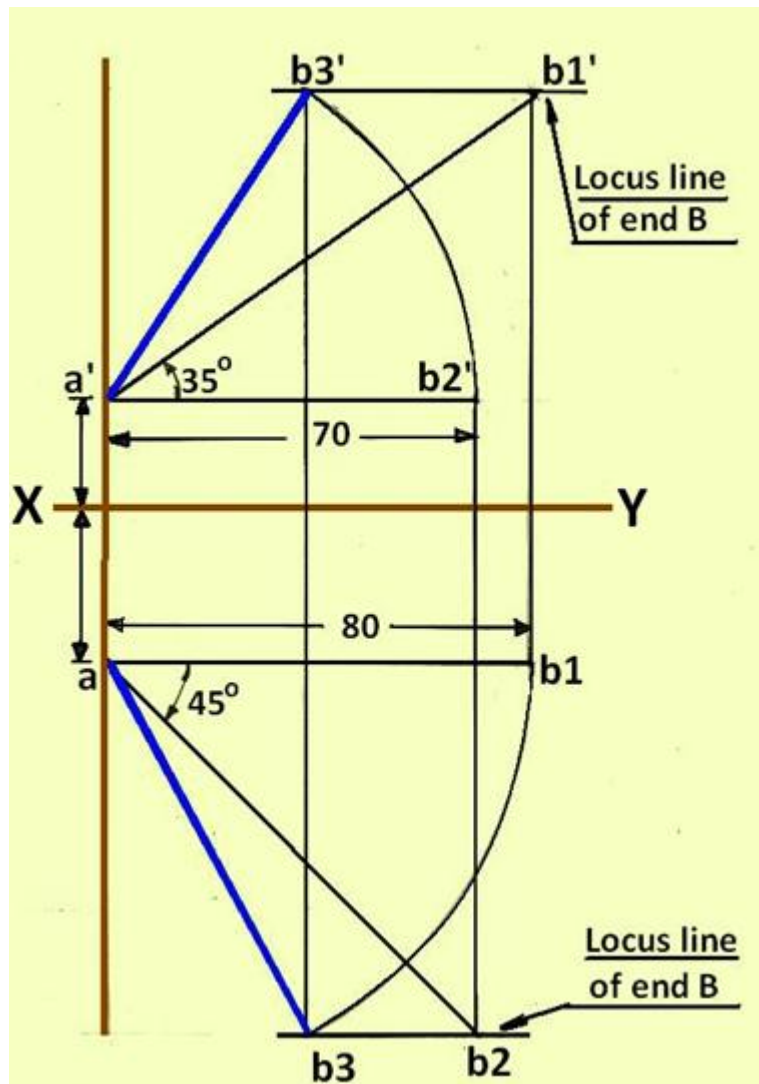


Figure 1. The projections of line AB in problem 1.

- Draw XY line and mark points a' (20 mm above XY line) and a (30mm below XY line) as given in the question.
- Let us assume that the line is parallel to HP and Inclined to VP.
- Draw a horizontal line from a' and mark distance equal to 70 mm on it. The end point is b2'. i.e., a' b2' = 70 mm.
- In this condition, the FV is parallel to XY line. TV can be obtained considering the following points:
- TV of point B will be along the vertical projector drawn from B2'.
- TV of line AB will be the true length = 100mm .
- The true inclination of the line with VP can also be obtained.
- Draw a projector downwards from b2'. It is clear that if TL of AB is drawn from a with the required inclination with the VP, it will give the distance a' b2'. Therefore, with radius equal to TL= 100mm and with centre a, cut an arc on the downward projector from b2'. Let this be ab2. Inclination of ab2 with horizontal will give true inclination of line AB with VP.
- Locus of point B is marked as a horizontal line at b2.
- Let us no assume that the line is parallel to VP and inclined to HP.
- In this condition, the TV is parallel to XY line. FV can be obtained considering the following points:
- FV of point B will be along the vertical projector drawn from B2'.

- FV of line AB will be the true length = 100mm .
- The true inclination of the line with HP can also be obtained.
- Draw a horizontal line from a and mark distance equal to 80 mm on it. The end point is b1. $a b_1 = 80$ mm.

Draw a projector upwards from b1. It is clear that if TL of AB is drawn from a' with the required inclination with the HP, it will give the distance ab1. Therefore, with radius equal to TL= 100mm and with centre a', cut an arc on the upward projector from b1. Let this be a'b1'. Inclination of a'b1' with horizontal will give true inclination of line AB with HP.

- Locus of point B is marked as a horizontal line at b1'. (1)

Drawing the top view and front view of line AB

1. The plan of AB (ab) is obtained as follows: With a as centre and radius equal to ab1, cut an arc on the locus line drawn at b2.
2. The elevation of AB (a'b') is obtained as follows: With a' as centre and radius equal to a'b2', cut an arc on the locus line drawn at b1'.

The required inclinations are: Angle of inclination with HP =350; Angle of inclination with VP = 450

Problem -2

Straight line AB is 40 mm long. End A is 10mm above HP and 15 mm in front of VP. FV of the line is inclined at 45° and TV is inclined at 60° to XY line. Draw the projections of line AB (FV and TV) and obtain the true inclination of line AB with HP and VP.

Solution: The solution for problem 2 is shown in figure 2. The step wise procedure for the solution is discussed below:

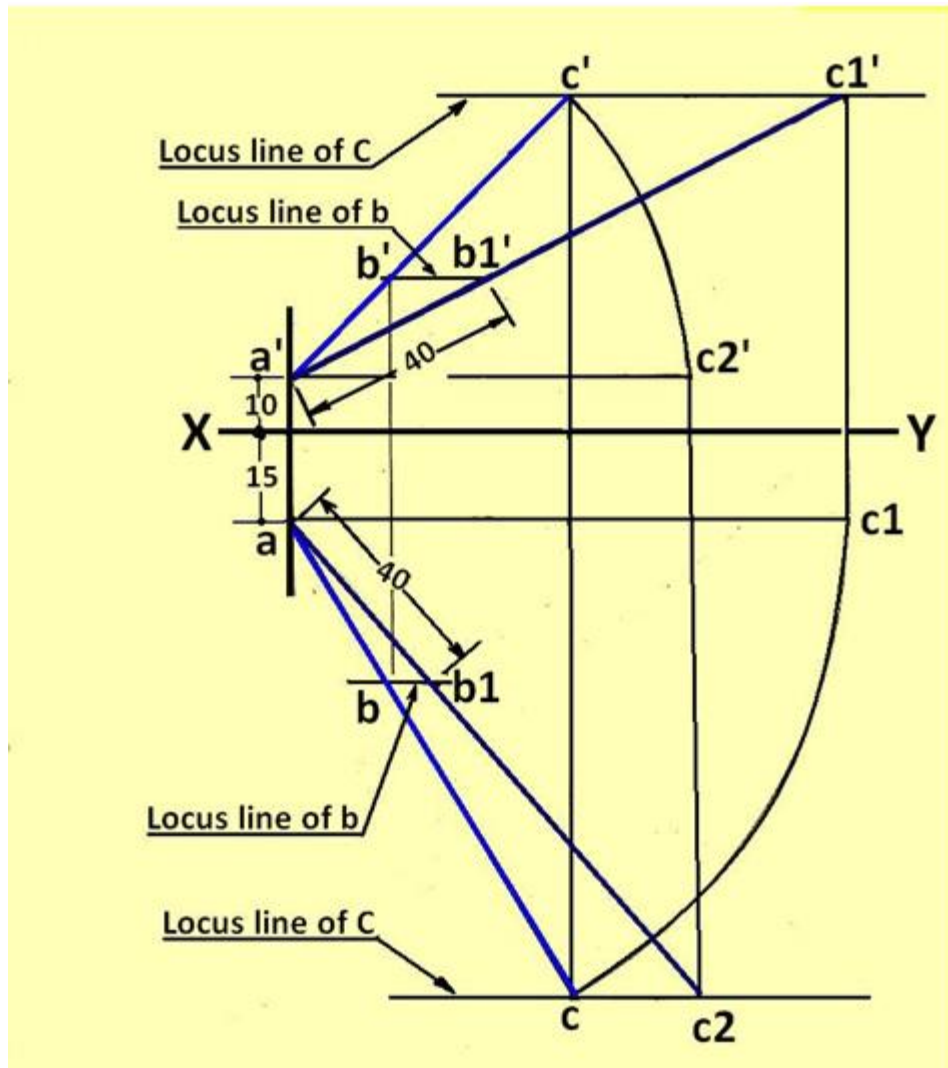


Figure 2. shows the solution of Problem 2.

- Mark the points a' and a according to the given data (i.e., 10 mm above and 15 mm below XY line, respectively).
- Since we don't know the exact position of point B in the FV, let us arbitrarily assume a point C on the line AB extended. Then the projections of points A, B, and C should line on the same line in all orientations. i.e the point corresponding to point b' will lie on line $a'c'$.
- Draw a line at 45° to horizontal from a' and mark any point c' on it.
- Under this situation, b' would lie on $a'c'$.
- From a draw a line at 60° to horizontal from a .
- Drop a projector from c' downwards to obtain the point c on 60° line.
- Draw horizontal lines at c and c' to denote the locus of point C.
- With a as centre and radius equal to ac , draw an arc to meet the horizontal through a at c_1 .
- Draw a projector upwards from c_1 to meet the horizontal locus line at c' . Let this meeting point be c_1' . $a'c_1'$ would then represent the TL and true inclination of AC with HP.
- On $a'c_1'$ mark b_1' such that $a'b_1'$ is equal to TL of $AB = 40\text{mm}$.
- Draw horizontal locus line through b_1' to meet $a'c'$ at b' .
- $a'b'$ is the required FV of AB. Drop projector downwards from b' and obtain TV ab .

- Similarly, with a' as centre and radius equal to $a'c'$, draw an arc to meet the horizontal through a' at $c2'$.
- Draw the projector downwards to meet the locus line at c at $c2$. $ac2$ would then represent the TL and true inclination of AC with VP.
- On $ac2$, mark $b1$ such that TL of AB (i.e. $ab1$) = 40 mm.
- Draw the locus line passing through $b1$ to line ac at b .
- ab is the required TV of line AB. The vertical projector through b' should pass through b .
- The required inclinations of line AB are: Angle of inclination with HP = 27° angle of inclination with VP = 50°

Worked out problem in Auxiliary projections

Problem 1.

The end projectors of a line AB is 40 mm. The point A is 24 mm above HP and 10 mm in front of VP. Point B is 46 mm above HP and 46 mm in front of VP. Determine the True length of the line and its True inclinations with both the reference planes.

Solution: The solution to the problem is shown in figure 1.

- Draw the XY line. Draw two lines perpendicular to XY line separated by a horizontal distance of 40 mm.
- Along the first line mark points a' and a 24 mm above and 10 mm below XY line.
- Along the second line mark b and b' 46 mm above and below XY line.
- Draw a reference line $X1 Y1$ parallel to line ab to represent the AVP.
- Through a and b , draw projectors to $X1 Y1$ line and extend it.
- Mark $a2'$ such that $3-a2' = 5-a'$. similarly mark $b2'$ such that $4-b2' = 6b'$.
- Join $a2'-b2'$ to obtain the true length of the line.
- The inclination of the line $a2'-b2'$ with $X1Y1$ line is the true inclination of the line with HP.
- Draw $X2Y2$ parallel to the front view $a'b'$ and project the auxiliary top view $a1b1$ in the similar manner. The inclination of the line $a1-b1$ with $X2Y2$ line is the true inclination of the line with VP.

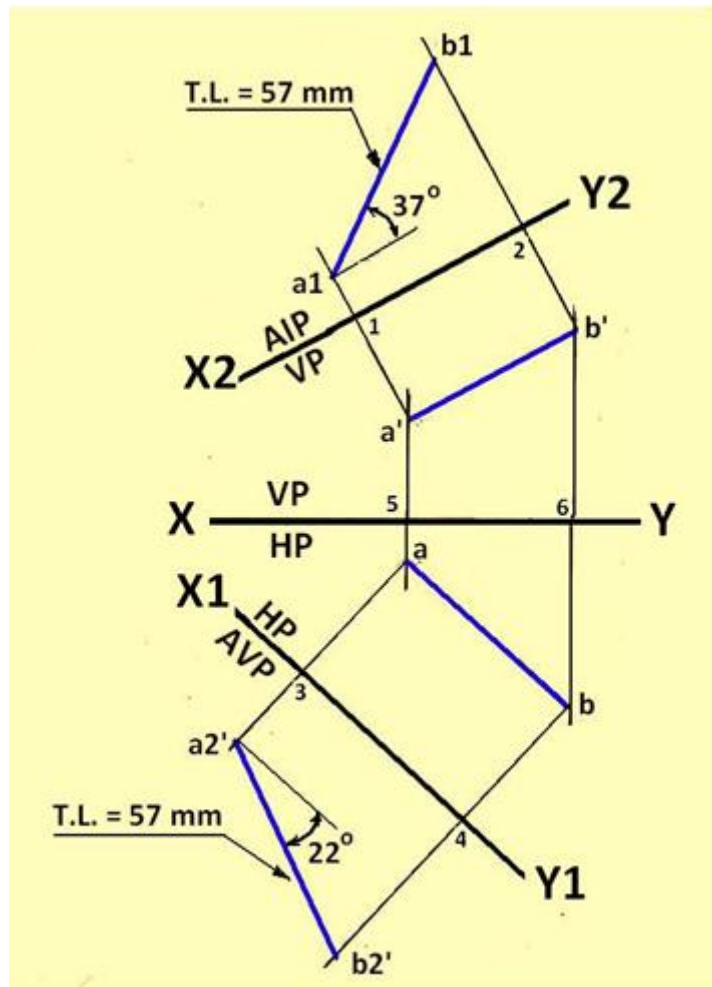


Figure1. Solution for problem No. 2.

By measurement, the following dimensions are obtained:

True length of the line is 57 mm; True inclination of the line with HP = 22° and True inclination of the line with VP = 37°

Problem 2.

A line AB 60 mm long has one of its extremities 60 mm in front of VP and 45 mm above HP. The line is inclined at 30° to HP and 45° to VP. Draw the projections of the line by the auxiliary plane method.

Solution: The solution to the problem is shown in figure 2.

- Draw lines $a1'-b1'$ and $a1-b1$ parallel and 45 mm above and 60 mm below XY lines respectively. This is the simple position when the line is parallel to both HP and VP.
- Draw line $X1Y1$ inclined at 30° to XY line representing the AIP.
- Draw projectors from $a1'$ and $b1'$ to line $X1Y1$ and extend it. Mark a and b equal to $a1-1$ and $b1-2$ from $X1Y1$ line to obtain the auxiliary front view.
- The following procedure is to be used to draw the auxiliary Top view:
- Since the line is also inclined at 45° to VP, The line $X2Y2$ will not be the apparent inclination. To obtain the apparent inclination of $X2Y2$, draw a line $ab2$ = true length of line $an(60\text{ mm})$ and inclined at 45° to ab .

- Draw the locus line passing through b2.
- With a as centre and radius equal to ab, draw an arc to intersect the locus line at b3.
- Inclination of line a-b3 with a-b is angle b, which will be the apparent inclination of X2Y2 with X1Y1.
- Draw X2Y2 inclined at b with X2Y1.
- Draw projectors from a and b to the AIP.
- Mark off a' and b' such that 5-a' = 3a1' and 6-b' = 4-b1'.
- a'-b' and a-b are the Top view and Front view of the line AB respectively.

Figure 2. Solution for problem No. 2.